

Thesis Proposal
**Explaining and Aligning Neural Models
for Machine Translation**

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Abstract

The recent success of large neural network models trained on massive amounts of unlabeled data has, yet again, highlighted the power of *scaling*, in terms of both model capacity and data quantity. However, despite its successes, scaling up also adds new additional challenges: (1) the data used to train these models is often noisy and contains harmful biases, often leading to models that do not **align** with their intended use case; (2) the large size of models makes it very difficult to understand how they work and to get interpretable **explanations** of their decisions.

This thesis addresses both challenges above, with a particular focus on models trained for **machine translation**. First, we start by showing that, by including metrics of *translation quality* (trained on human feedback annotations) in the inference stage of a trained MT model, we can improve the quality of the translation and align the system towards *human-like* translations. We then also show that we can almost completely remove the dependency on human annotations by leveraging *large language models* to provide (fine-grained) translation quality feedback. Then, we propose a novel method for extracting *post-hoc* explanations from trained neural network models, relying on the assumption that good explanations should help other models learn and leveraging bilevel optimization/meta-learning to learn explainers that teach well.

Finally, based on the insight that both *feedback* and *explanations* have similar goals, we propose unifying both through *AI Teachers*, and comparing the efficacy of using feedback and explanations (as well as *synthetic data*) to improve the quality of machine translation models, by leveraging LLMs to provide them at comparable costs. Following the insights of this comparison, we propose a novel framework for sample-efficient training and alignment of MT and language models: we train multiple generations of models, where each generation leverages feedback, explanations, and/or synthetic data from the previous generation to improve its performance. Importantly, we plan on studying the *scaling* of each approach to leverage the teachers’ knowledge since, as the proficiency of the AI teachers increases over generations, their evolving capabilities could potentially alter the effectiveness of each approach.

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List of Algorithms

Introduction

The field of natural language processing (NLP) has seen explosive growth in interest and successful applications in recent years, largely driven by the efficacy of parametric models of language. In particular, massive neural network models (often with billions of parameters) leveraging massive amounts of textual data from the internet (frequently surpassing trillions of words) have been able to match or even surpass human performance on a variety of natural language tasks. In machine translation (MT), for example, state-of-the-art systems are now able to translate between *high-resource* languages (e.g. English, German, etc.) with near-human quality [Koc+22].

However, despite the overall success of *scaling* (both in terms of model capacity and training data) at improving the general capabilities of these models, it also adds new additional challenges to their widespread adoption (besides the computational ones):

- The amount of data required to train these models is so large that manually-curated data is insufficient, and they have to be trained on data “scraped” from the internet, loosely filtered with automated procedures. This leads to models that are often trained on noisy data, potentially containing harmful biases, leading to models that don’t completely *align* with their intended use case.
- The size of these models, coupled with the nature of neural network computations, makes it near-impossible to get a mechanistic understanding of how they work, and getting interpretable and “*faithful*” non-mechanistic explanations of their decisions is still an open problem. This makes it difficult to understand how to improve them, and to trust them in safety-critical applications.

In this thesis, we aim to contribute techniques to tackle both these challenges and to show that we can train *machine translation* (MT) models to generate more human-like translations while using less *human-annotated* data and producing interpretable explanations of their translation process. Importantly, we hypothesise that both these challenges can be tackled under a single lens (with insights from one helping the other), and that both can be addressed by leveraging Large Language Models (LLMs) as *AI Teachers*.

1.1 Thesis Overview

We start by providing a brief overview of the background and related work for the topics covered in this thesis in [Chapter 2](#). The rest of the thesis is then divided into three parts:

Part I: [Aligning Machine Translation with Feedback](#) Firstly ([Chapter 3](#)), we explore how we can improve the quality of MT models, trained of large corpora of (noisy) translation-pairs, by modifying the decoding algorithm to not only consider model-likeness but also metrics of translation quality, effectively aligning the model with the goal of *human-like translation*. To do so, we propose incorporating recent reference-based and reference-free *neural, trained* metrics (trained to align with human judgement of translation quality) into the decoding algorithm through techniques such as *N*-best reranking and *minimum Bayes risk* decoding. We show that models that leverage these metrics can produce more human-like translations (even if traditional automatic metrics would disagree). After ([Chapter 4](#)), we also show that LLMs can be leveraged as state-of-the-art *fine-grained* translation quality metrics, with only a few supervised examples (in contrast to previous trained neural metrics which require many orders of magnitude more data), and hinting at the possibility of using LLMs to align machine translation models.

Part II: [Model Explanations as Learning Signals](#) We then move onto the problem of *interpretability*, and how to extract *explanations* from neural network models. While many definitions of what is a good explanation have been proposed, we focus on a quantitatively measurable definition based on the assumption that good explanations should help other models *learn*. We propose a novel method for extracting *post-hoc* explanations from trained neural network models, leveraging bilevel optimization/meta-learning to learn (parametrized) explainers that help teach student models ([Chapter 5](#)). We show not only that our method is able to extract explanations with higher *simulability* (i.e. that help students simulate the teacher) than other methods, but that this explanations are more *plausible*, aligning with human explanations.

Part III: [Learning with AI Teachers \(Proposal\)](#) In this proposal, we explore the hypothesis that, when viewing *explanations* as learning signals, they can be related and compared to *feedback* if the former is optimized for and evaluated against same goal behaviour as the latter. In particular, considering the recent success of LLM-provided *AI feedback* at aligning and improving the performance of generation models, and ours (and others) success extracting and leveraging *fine-grained* feedback from the same models, we propose generalizing them as *AI teachers* ([Chapter 6](#)), and comparing the efficacy of various *learning signals* they can provide (feedback, explanations or even and synthetic data) to improve the quality of machine translation models. Importantly, we plan on studying not only the impact of *learning signal* from larger teachers but also if it is possible to leverage equivalent or even smaller teachers to improve the quality of models. Then ([Chapter 7](#)), informed by our finding, we propose exploring the possibility of training *multi-generational* LLMs, which learn using a combination of unlabelled data and *learnings* from previous

generations. We hypothesize that if the choice of *learning* used at each generation is empirically informed by the (evolving) capabilities of models, we can train more aligned and interpretable models for tasks like machine translation in a more sample-efficient manner.

1.2 Summary of Publications

Parts of the background and related work in [Chapter 2](#) appear in the survey:

[Fer+23a] Patrick Fernandes et al. “Bridging the Gap: A Survey on Integrating (Human) Feedback for Natural Language Generation”. In: *Transactions of the Association for Computational Linguistics* (2023). URL: <https://arxiv.org/abs/2305.00955>

The content of [Chapter 3](#) appears in:

[Fer+22b] Patrick Fernandes et al. “Quality-Aware Decoding for Neural Machine Translation”. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Marine Carpuat et al. Seattle, United States: Association for Computational Linguistics, July 2022, pp. 1396–1412. URL: <https://aclanthology.org/2022.naacl-main.100>

The content of [Chapter 4](#) appears in:

[Fer+23c] Patrick Fernandes et al. “The Devil is in the Errors: Leveraging Large Language Models for Fine-grained Machine Translation Evaluation”. In: *Proceedings of the Seventh Conference on Machine Translation (WMT)*. Association for Computational Linguistics, Dec. 2023

The content of [Chapter 5](#) appears in:

[Fer+22a] Patrick Fernandes et al. “Learning to Scaffold: Optimizing Model Explanations for Teaching”. In: *Advances in Neural Information Processing Systems*. 2022. URL: <https://openreview.net/pdf?id=V5rlSPsHpkf>

Non-thesis research I have also pursued other research directions, which have lead to the following first-author publications (these are excluded for the remainder of this thesis):

[Fer+21] Patrick Fernandes et al. “Measuring and Increasing Context Usage in Context-Aware Machine Translation”. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Ed. by Chengqing Zong et al. Online: Association for Computational Linguistics, Aug. 2021, pp. 6467–6478. DOI: [10.18653/v1/2021.acl-long.505](https://doi.org/10.18653/v1/2021.acl-long.505). URL: <https://aclanthology.org/2021.acl-long.505>

[Fer+23d] Patrick Fernandes et al. “When Does Translation Require Context? A Data-driven, Multilingual Exploration”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Anna Rogers et al. Awarded **Best Resource Paper**. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 606–626. DOI: [10.18653/v1/2023.acl-long.36](https://doi.org/10.18653/v1/2023.acl-long.36). URL: <https://aclanthology.org/2023.acl-long.36>

[Fer+23b] Patrick Fernandes et al. “Scaling Laws for Multilingual Neural Machine Translation”. In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 23–29 Jul 2023, pp. 10053–10071. URL: <https://proceedings.mlr.press/v202/fernandes23a.html>

Additionally, I have also contributed to the following publications as a non-first author:

[Yin+21] Kayo Yin et al. “Do Context-Aware Translation Models Pay the Right Attention?” In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Ed. by Chengqing Zong et al. Online: Association for Computational Linguistics, Aug. 2021, pp. 788–801. DOI: [10.18653/v1/2021.acl-long.65](https://doi.org/10.18653/v1/2021.acl-long.65). URL: <https://aclanthology.org/2021.acl-long.65>

[Tre+22] Marcos Treviso et al. “Predicting Attention Sparsity in Transformers”. In: *Proceedings of the Sixth Workshop on Structured Prediction for NLP*. ed. by Andreas Vlachos et al. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 67–81. DOI: [10.18653/v1/2022.spnlp-1.7](https://doi.org/10.18653/v1/2022.spnlp-1.7). URL: <https://aclanthology.org/2022.spnlp-1.7>

[Yan+22] Brian Yan et al. “CMU’s IWSLT 2022 Dialect Speech Translation System”. In: *Proceedings of the 19th International Conference on Spoken Language Translation (IWSLT 2022)*. Ed. by Elizabeth Salesky et al. Dublin, Ireland (in-person and online): Association for Computational Linguistics, May 2022, pp. 298–307. DOI: [10.18653/v1/2022.iwslt-1.27](https://doi.org/10.18653/v1/2022.iwslt-1.27). URL: <https://aclanthology.org/2022.iwslt-1.27>

[Sun+23] Jimin Sun et al. “A Multi-dimensional Evaluation of Tokenizer-free Multilingual Pretrained Models”. In: *Findings of the Association for Computational Linguistics: EACL 2023*. Ed. by Andreas Vlachos and Isabelle Augenstein. Dubrovnik, Croatia: Association for Computational Linguistics, May 2023, pp. 1725–1735. DOI: [10.18653/v1/2023.findings-eacl.128](https://doi.org/10.18653/v1/2023.findings-eacl.128). URL: <https://aclanthology.org/2023.findings-eacl.128>

[Far+22] Ana C Farinha et al. “Findings of the WMT 2022 Shared Task on Chat Translation”. In: *Proceedings of the Seventh Conference on Machine Translation (WMT)*. ed. by Philipp Koehn et al. Abu Dhabi, United Arab Emirates (Hybrid): Association for Computational Linguistics, Dec. 2022, pp. 724–743. URL: <https://aclanthology.org/2022.wmt-1.70>

[Yan+23] Brian Yan et al. “ESPnet-ST-v2: Multipurpose Spoken Language Translation Toolkit”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*. Ed. by Danushka Bollegala et al. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 400–411. DOI: [10.18653/v1/2023.acl-demo.38](https://doi.org/10.18653/v1/2023.acl-demo.38). URL: <https://aclanthology.org/2023.acl-demo.38>

[Jai+23a] Sameer Jain et al. “Multi-Dimensional Evaluation of Text Summarization with In-Context Learning”. In: *Findings of the Association for Computational Linguistics: ACL 2023*. Ed. by Anna Rogers et al. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 8487–8495. DOI: [10.18653/v1/2023.findings-acl.537](https://doi.org/10.18653/v1/2023.findings-acl.537). URL: <https://aclanthology.org/2023.findings-acl.537>

[HFZ23] Sumire Honda et al. “Context-aware Neural Machine Translation for English-Japanese Business Scene Dialogues”. In: *In Proceedings of Machine Translation Summit XIX, Vol. 1: Research Track* (2023)

And the following pre-prints (which are currently planned for/under submission):

[Ram+23] Miguel Moura Ramos et al. *Aligning Neural Machine Translation Models: Human Feedback in Training and Inference*. 2023. arXiv: [2311.09132](https://arxiv.org/abs/2311.09132) [cs.CL]

[Fay+24] Manuel Faysse et al. *CroissantLLM: A Truly Bilingual French-English Language Model*. 2024. arXiv: [2402.00786](https://arxiv.org/abs/2402.00786) [cs.CL]

[Alv+24] Duarte M. Alves et al. *Tower: An Open Multilingual Large Language Model for Translation-Related Tasks*. 2024. arXiv: [2402.17733](https://arxiv.org/abs/2402.17733) [cs.CL]

1.3 Proposed Timeline

We plan on concluding thesis over the next year and a half, with the following timeline:

+	March 2024	Thesis Proposal
+	April 2024 - Sept. 2024	Conclude Chapter 6
+	Oct. 2024 - March 2025	Conclude Chapter 7
▼	July 2025	Thesis Defense

Background

In this chapter, we provide an overview of the main concepts necessary to understand and motivate the remainder of this thesis. We start by introducing the (conditional) language modelling task and neural language models (looking at neural machine translation as a specific case) and discuss how *scaling* (of these models) has underpinned much of the recent success in NLP (Section 2.1). We then discuss how *feedback* can be used to align and further improve the performance of neural language models (Section 2.2). Finally, we discuss the *interpretability* problem with neural models (Section 2.3), and one approach to define and quantitatively measure how *good* an explanation of a model’s decision is.

2.1 Learning Parametric Models of Language

Language modelling is the task of estimating the probability of sequences of words or symbols occurring in a language of interest. We might want, for example, to estimate the probability of a sentence appearing in the English language, or the probability of a sequence of notes appearing in a piece of music.

Formally, given a sequence of tokens $y = (y_1, \dots, y_T)$ from some vocabulary $y_i \in \mathcal{V}$, the goal is to estimate the probability of the sequence $p(y) = p(y_1, \dots, y_T)$. To simplify the task, it is common to *autoregressively* factorise the probability of a sequence as the product of the probabilities of each token in the sequence *conditioned on the previous ones*, i.e.

$$p(y_1, \dots, y_T) = \prod_{i=1}^T p(y_i \mid y_{i-1}, \dots, y_1) \quad (2.1)$$

To tractably solve this task, one can estimate each conditional probability using a *parametric* model $p_\theta(y_i \mid y_{i-1}, \dots, y_1)$, with parameters θ . The problem then reduces to choosing a parametric model that is expressive enough to capture the distribution of interest, and to estimate its parameters θ *from data*.

The latter is often tackled through an optimisation perspective: given some training samples from language $\mathcal{D} = \{y^{(i)}\}_{i=1}^N$, the goal is to find the parameters θ^* that maximise

the likelihood of language samples or, equivalently, minimise the *negative log-likelihood* of the data:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \mathcal{L}(\theta) \quad (2.2)$$

$$\mathcal{L}(\theta) = \mathbb{E}_{y \sim \mathcal{D}} \left[- \sum_{i=1}^T \log p_{\theta}(y_i | y_{<i}) \right] \quad (2.3)$$

Once the parameters θ^* are estimated, language models can be used to, not only estimate the probability of a sequence but also generate new sequences in the language by, for example, sampling from the model distribution $\hat{y}_i \sim p_{\theta^*}(y_i | \hat{y}_{i-1}, \dots, \hat{y})$. Additionally, language modelling can be extended to the *conditional* case, where the goal is to estimate the probability $p(y | x)$ of a sequence y given another sequence x (from the same or another language).

While conceptually a simple task, most problems in NLP can be framed as (conditional) language modelling [Raf+20a]. Language models can be evaluated through their *perplexity* of some unseen data but, in practice, the goal is often to use language models to solve other tasks and, as such, task-specific metrics are generally used to evaluate their performance.

2.1.1 Neural Language Models

The choice of the parametric model plays an important role in the difficulty and approach to solving Equation 2.2. While early approaches to language modelling relied on *count-based* models, such as n -gram models, the current dominant approach is to use *neural networks* to parameterise the conditional (token) probabilities [Ben+03].

Formally, a neural language model parameterises the conditional probability of tokens at each step as

$$p_{\theta}(y_i | y_{i-1}, \dots, y_1) = \text{NORMALISE}(f_{\theta}(y_1, \dots, y_{i-1})), \quad (2.4)$$

where $f_{\theta} : \mathcal{V}^* \rightarrow \mathbb{R}^{|\mathcal{V}|}$ denotes a neural network model parameterised by θ that returns scores or *logits* for each next possible tokens, and $\text{NORMALIZE} : \mathbb{R}^{|\mathcal{V}|} \rightarrow \Delta^{|\mathcal{V}|-1}$ normalises these scores into a probability distribution, typically using a **SOFTMAX** function:

$$\text{SOFTMAX}(z)_i = \frac{\exp(z_i)}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_j)}. \quad (2.5)$$

The core advantage of using neural networks is that, by defining a suitable loss function, the parameters θ can be tractably estimated with *stochastic gradient descent*, with gradients computed using the *backpropagation* algorithm [RHW86]. In the case of neural language models, this loss is precisely the negative log-likelihood of the data, as defined in Equation 2.3.

Notice that, since sentences in a language can have different lengths, the neural network model f_{θ} must handle variable-length sequences, something standard *feedforward* neural networks are not capable of. While *recurrent* neural networks (RNNs) [HS97] have traditionally been used for this purpose, the current state-of-the-art neural language models generally use attention-based models, such as Transformers [Vas+17a].

2.1.2 Neural Machine Translation

Machine Translation (MT) is the task of translating a sentence or document from a source language to a target language, intending to convey the same meaning as closely as possible. While earlier approaches to MT used rule-based systems and statistical models, the currently dominant approach is to frame MT as a *conditional language modelling* task, and use neural language models to solve the task [SVL14]. This is known as neural machine translation (NMT).

Formally, given a dataset of sentence pairs $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$, where $x^{(i)}$ and $y^{(i)}$ are sentences in the source and target languages respectively, NMT models are trained to minimise the *conditional* negative log-likelihood of the target sentences given the source sentences:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[- \sum_i \log p_{\theta}(y_i | y_{<i}, x) \right] \quad (2.6)$$

Once an NMT model is trained, one typically obtains translations from it by attempting to find the *most likely* translation given a source sentence x , i.e. $\hat{y} = \operatorname{argmax}_y p_{\theta}(y|x)$ ¹. Since this problem is intractable, one typically uses approximate search algorithms, such as *beam search* [Gra12b].

The gold standard to evaluate the quality of NMT systems is to use *human evaluation*, where bilingual (professional) translators compare and score the system’s translated sentences according to how well they convey the same meaning as the original source sentence [Fre+21a]. However, this is impractical for routine evaluation, and automatic metrics are the most common approach to evaluate machine translation. While early MT evaluation relied predominantly on *lexical* metrics, such as BLEU [Pap+02a], recent work has shown that *neural* metrics trained to correlate with human judgments of translation quality, such as COMET [Rei+20a] and BLEURT [SDP20], are more effective at capturing the quality of translations [Fre+22].

2.1.3 Scaling & Transfer

For much of NLP’s history, the typical paradigm was to treat each task as a separate problem, with dedicated datasets and specific models designed, or at least separately trained, for each task. Moreover, these models were relatively small (very rarely surpassing 1 billion parameters), as increasing the model size was generally believed to lead to *overfitting* to the task-specific training dataset.

However, in recent years the paradigm has shifted towards training large *pretrained* models with a language modelling objective on a large corpus of text (generally from the internet) since these models tend to perform well on a variety of tasks after a subsequent fine-tuning stage on small amounts of task-specific training data [RN18], or even without any additional training or data [Bro+20]!

¹While in MT this decoding objective tends to obtain higher-quality translations, it also possible decode sampling from the distribution of possible translations, like in other language modelling tasks (§3)

Underpinning this shift are two empirical observations that have proven key in the success of pretrained models:

- **Scaling**, either by increasing the model size or the amount of training data, is one of the most successful approaches to improve the performance of neural (language) models, and the increase in performance through scaling is predictable from smaller models [Kap+20].
- **Transfer**, where neural language models that achieve good performance (low perplexity) on modelling a specific language (i.e English) tend to learn “universal representations” of that language. When these representations are leveraged through *transfer learning* to solve other tasks, they lead to models that are more sample-efficient and perform better than models trained from scratch [Raf+20b], and they can even be used directly through *zero-shot prompting* or *in-context learning* [Bro+20]

These observations have recently culminated in the development *large language models* (LLMs), models with billions of parameters trained on internet-scale data, which have shown remarkable performance across a wide range of NLP tasks [Ope23], including generating high-quality translations [Vil+22]. However, another ingredient was needed to leverage the full potential of these models.

2.2 Aligning Neural Models with Feedback

While the scaling and transfer have been key to recent breakthroughs in NLP, they also brought new challenges that task-specific models did not have to deal with. To leverage the benefits of scale, these models are trained to maximize the likelihood (Equation 2.2) of large amounts of (unlabeled) data in a language. However, the data used to train these models is generally *scraped* from the Internet, often containing noise, social biases, and errors [Bol+16; Dod+21], which can result in a *misspecification* of target behaviour [Ken+21]. This may lead to models that generate toxic, inaccurate, and unhelpful content [She+19; Ben+21], issues that might persist even after task-specific fine-tuning of pretrained models.

To address this, recent works have proposed moving from the traditional paradigm of static supervised learning, and instead also incorporating **human feedback** on model-generated outputs during the training or inference process. This move has led to the emergence of the first widely-used general-purpose language assistants [Ope23].

2.2.1 Defining Human Feedback

Formally, we consider **human feedback** to be a family of functions $\mathcal{H}$ such that $h \in \mathcal{H}$ takes an input $x \in \mathcal{X}$ and one or more outputs $y_1, \dots, y_n \in \mathcal{Y}$ (for example, by sampling from a trained language model) and returns some *feedback* $f \in \mathcal{F}$:

$$h : \mathcal{X} \times \underbrace{\mathcal{Y}_1 \times \dots \times \mathcal{Y}_n}_n \rightarrow \mathcal{F}. \quad (2.7)$$

A simple example of a feedback function is asking humans to say if, given an input, a particular output is good or bad ($h : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$). However, more complex feedback

Input	Output(s)	Feedback	Type
		0.7	Score
<i>A melhor comida do mundo é a portuguesa.</i>	<i>The worst food in the world are Portuguese.</i>	'worst': major/accuracy 'are': minor/fluency	MQM
		'worst' → 'best', 'are' → 'is'	Post-Editon
<i>Artificial intelligence has the potential to revolutionize industries (...) but ethical concerns need to be handled.</i>	<i>AI can change industries.</i>	Fluency: 1 Relevance: 0.7	Multi-Aspect
		"Misses the ethical concerns."	Natural Language
<i>Explain the moon landing to a 6 year old</i>	A: <i>People went to the ...</i> B: <i>The moon is a satellite...</i>	A > B	Ranking

Table 2.1: Example input and output for three tasks (machine translation, summarization, and instruction following) and possible different (example) feedback that can be given.

functions, such as rankings or natural language feedback, are also commonly used (see Table 2.1 for examples). The choice of format has implications on the expressivity of the feedback, the ease of its collection, and how we can use it to improve systems, and the level of *rationality* of said feedback is heavily impacted by this choice [Gho+23].

This feedback is collected to *align* the model’s behaviour with some (often ill-defined) *goal* behaviour: for example, we might want our summarization model to generate summaries that contain all core information, even if it means they are longer; or in *commercial* machine translation, where we want to ensure that models do not mistranslate business-critical information and have good *translation quality*. This **alignment objective** has been studied extensively in the *AI safety and alignment* literature [Bos14; Amo+16; Ken+21], but can be broadly framed and studied according to two sub-objectives and how they interact: increasing the model’s *helpfulness* and/or its *harmlessness* [Bai+22b].

2.2.2 Leveraging Feedback to Improve Generation

In an ideal scenario, we would directly leverage human feedback to improve generation. One way to use it is by optimizing the model parameters directly. This requires the feedback to be “optimizable”, *i.e.*, possibly formulated as an optimization problem based on which we can obtain an improved model. For instance, if the feedback is a numerical preference score ($f \in \mathbb{R}$), we can create the following optimization problem:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \mathbb{E}_{x \sim \mathcal{D}} [h(x, \hat{y})] \quad (2.8)$$

where $\mathcal{D}$ is the distribution of possible inputs, and $\hat{y}$ is an output generated by the model.

Various techniques have been suggested to optimize θ using the collected human feedback. The most popular approach is to use *reinforcement learning* (RL), where the feedback is used as a reward signal, and use common RL algorithms such as REINFORCE [Wil92], which updates the parameters with the gradients. However, alternatives to RL exist, such as jointly modelling the feedback and language from a dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)}, f^{(i)})\}_{i=1}^N$, with $f^{(i)} = h(x^{(i)}, y^{(i)})$:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y,f) \sim \mathcal{D}} [-\log p_{\theta}(f, y \mid x)] \quad (2.9)$$

and factorizing the joint probability (by predicting the feedback after generating an output or predicting an output for a given specific feedback).

2.2.3 Learning Models of Human Feedback

Directly using human feedback to improve model behaviour is not feasible in the general case: asking humans to provide feedback for *every* model output is both expensive and time-consuming. An alternate approach to obtaining human feedback is to develop models that can predict or approximate it. Although they may not be perfect, they can provide feedback at a low cost after training, enabling feedback-dependent techniques at scale.

More formally, given a feedback function $h : \mathcal{X} \times \mathcal{Y}_1 \times \dots \times \mathcal{Y}_n \rightarrow \mathcal{F}$, we want to learn a *parametric* (numerical) feedback model $\hat{h}_\phi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ (with parameters ϕ) that “agrees” with human feedback. This agreement is expressed through a loss which the model is trained to minimize.

$$\phi_\star = \operatorname{argmin}_{\phi} \mathbb{E}_{x, y_1, \dots, y_n \sim \mathcal{D}_f} [\mathcal{L}(\phi)] \quad (2.10)$$

$$\mathcal{L}(\phi) = \operatorname{loss} \left(\hat{h}_\phi(x, y_1), \dots, h(x, y_{1:n}) \right) \quad (2.11)$$

For example, if the feedback function we are trying to model is also numerical ($h : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$), then this loss can just be any standard regression loss, such as the squared difference between the human feedback and model feedback

$$\mathcal{L}(\phi) = \left(\hat{h}_\phi(x, y) - h(x, y) \right)^2 \quad (2.12)$$

Feedback modelling has been studied extensively in the context of *learning evaluation metrics* for NLP. For example, in MT, BLEURT [SDP20] and COMET [Rei+20a] were trained to regress on human translation quality assessments (using a loss similar to Equation 2.12) and, as mentioned, this has led them to be much more effective than traditional lexical metrics². Notably, these metrics were not trained with the intent of improving generation directly, though they can be used for this purpose (as we will show in Chapter 3). More recently, there has been more interest in developing feedback models directly for improving generation [Böh+19; Zie+19].

After training a feedback model, we can use it almost exactly as we would use human feedback, for example, by optimizing the model parameters with feedback it gives³. Additionally, using a feedback model also opens the door to leveraging feedback during the decoding process, which can be used to improve the performance of an already trained model (as we will show in Chapter 3).

²Reference-based MT metrics can be seen feedback models which take as an additional input a *reference* translation, which eases the difficulty in approximating translation feedback

³Since feedback models are imperfect proxies of human feedback, generally regularization is added when training with them to avoid “reward overfitting” [GSH22]

2.2.4 AI Feedback

Feedback models have been crucial in advancing generation techniques by effectively leveraging feedback. However, they are heavily reliant on human input: for example, [Gao et al. \[GSH22\]](#) found that across various preference model sizes, utilizing fewer than 1,000 comparisons resulted in only chance improvements. Moreover, employing static feedback can make consistency challenging, causing changes in the model’s output distribution.

AI-generated feedback, an emerging research area, focuses on harnessing the LLM’s own abilities to enhance the model without human intervention. For example, LLMs can be prompted to assess the harmfulness of their outputs and revise them accordingly [[Bai+22a](#)], or an external LLM can be used to critique the task model’s outputs [[Mad+21](#); [Aky+23](#)]. Generally, larger LLMs provide feedback closer to humans and are particularly effective at generating feedback when they themselves have been trained to adhere to align with human preferences.

While AI-generated feedback still hasn’t replaced human feedback (which is implicitly used to train the LLM and design the *prompt* that generates feedback), it has the potential to significantly reduce the need for human feedback when training and aligning *new* models, and generality of LLMs allow us to explore a wide range of feedback types and formats, providing a good model to study the impact of these choices on alignment at scale (as we will discuss in [Chapter 6](#)).

2.3 Interpreting Neural Network’s Decisions

The widespread adoption of neural networks across fields has been accompanied by increasing demand for understanding how these models work and to provide *explanations* for particular decisions and generations they make [[Lip16b](#)]. However, the complex and highly nonlinear nature of neural network models makes it almost impossible to get a “causal” understanding of how inputs to the model lead to outputs, and the challenge is exacerbated with large models.

Therefore, researchers have developed various methods for extracting *post-hoc* explanations from trained neural network models, which can be used to provide insights into how the model works and to provide explanations for particular decisions.

2.3.1 Extracting Local Explanations

What exactly is an “explanation” for a particular depends on the specific task, application and audience. However, in general, we are interested in understanding how the model uses its input to make a decision or generate output. In particular, *local* explanations aim to understand and interpret individual predictions or outputs, as opposed to *global* explanations which aim to reveal insights about the model’s overall behaviour.

Feature Attribution One common approach to extracting local explanations from neural models is through *feature attribution* methods. These methods aim to distribute the

importance of a prediction over the input features (for example, words in a sentence or pixels in an image) to get an intuition of which parts of the input have contributed to the decision or output.

A popular method for feature attribution is the *gradient-based* method which attributes the importance of each input feature based on the gradient of the output concerning that input feature. Formally, given a model f_θ that for an input $x = (x_1, \dots, x_n)$ outputs a prediction $f_\theta(x)$ the gradient-based *score*

$$\mathbf{g}(x_i) = \|\nabla_{x_i} f_\theta(x)\|_2 \quad (2.13)$$

is a measure of how changing the feature x_i affects the output $f_\theta(x)$, and can be interpreted as the importance of that feature to the final prediction. Many variations of these gradient-based methods have been proposed [DDF14; STY17a], and they are commonly used to provide explanations since they can be used with any differentiable (neural network) model.

Alternatively, the prevalence of transformer-based models in NLP has led to feature attribution methods that leverage the attention mechanism in these models to provide explanations. Each attention mechanism naturally relies on computing, for each input/output feature x_i , distribution of scores over all input features that input “attends” to

$$\alpha_i = \text{SOFTMAX}(q_i K) \quad (2.14)$$

where $q_i \in \mathbb{R}^d$ is a query vector for the output feature x_i , and $K \in \mathbb{R}^{n \times d}$ is a matrix of key vectors for the input features. These attention scores can then be aggregated to provide an importance score for each input feature, for example, by picking a relevant input feature (for example, the first or last token in a sentence) and summing or averaging the attention scores produced in all the attention heads or from a given layer [Vas+19; WP19].

Chain-of-Thought One interesting property of LLMs is that their performance on downstream tasks can generally be improved by prompting these models to generate a series of intermediate reasoning steps (“*Let’s think step-by-step*”) [Wei+22]. While this *chain-of-thought* phenomenon is not fully understood and has been primarily used to improve model performance, the intermediate generations can be interpreted as *free-form* explanations of the model’s decision and can be representative of the model’s underlying decision-making process [Lan+23].

2.3.2 What Makes a Good Explanation?

While there are many methods for extracting explanations from neural models, the question of what method provides better or worse explanations and which one to use is not straightforward. This is exacerbated by the fact that, often, different methods produce explanations that disagree with each other for the same model and input [Nee+21].

Commonly, the quality of an explanation is evaluated according to its *plausibility*, by comparing it to some *ground truth* explanation provided by a human (given the same input and output). However, this assumes that the model’s decision process should always match the human process, an assumption that can lead to *unfaithful* explanations that don’t

represent the model’s actual decision process [Pru+20b]. Instead, alternative quantitative measures of explanation quality that don’t rely on human-provided explanations have been proposed [DeY+20; Tre+22].

In particular, Pruthi et al. [Pru+22] proposed a *simulability*-based framework to evaluate the quality of an explanation: given a *teacher* model and explanations for its decisions, a *student* model S with parameters θ_S is trained to predict the teacher’s decisions using *and* its explanations

$$\mathcal{L}_S(\theta_S) = \mathbb{E}_{(x,y,e) \sim \mathcal{D}} [-\log p_{\theta_S}(e, y \mid x)] \quad (2.15)$$

where $\mathcal{D}$ is a dataset of input and the teacher’s decisions and explanations, and ϕ_S are the parameters of the student model. The quality of the explanation is then evaluated by the performance of the student model at predicting the teacher’s decisions on some held-out data, and compared to a student model that is trained without explanations.

Intuitively, this framework assumes that a *good* explanation of the model’s underlying decision process should help the student model learn to simulate the teacher’s decisions better than a model that is trained without explanations. This instrumentalist assumption brings explanations much closer to *feedback* in how it can be treated since they have similar goals (notice the similarities between the previous equation and Equation 2.9; we will discuss these further in Chapter 6).

Part I

Aligning Machine Translation with Feedback

Quality-Aware Decoding for Neural MT

In this chapter, we introduce **quality-aware decoding**, a framework for decoding in neural machine translation (NMT). We leverage recent neural metrics for MT evaluation trained to approximate human feedback (§2.2.3) to rank translation candidates produced by NMT models, effectively aligning the model with the goal of *human-like translation*. We show that quality-aware decoding leads to higher-quality translations than traditional decoding methods, both according to automatic metrics and human evaluation.

The code for the experiments in this chapter is available open-source at github.com/deep-spin/qaware-decode.

3.1 Introduction

The most common procedure in neural machine translation (NMT) is to train models using maximum likelihood estimation (MLE) at training time (Equation 2.6), and to decode with beam search at test time, as a way to approximate maximum-a-posteriori (MAP) decoding. However, several works have questioned the utility of model likelihood as a good proxy for translation quality [KK17; Ott+18; SB19; EA20], with the most probable translation of an MLE-trained model often having poor quality¹. In parallel, significant progress has been made in methods for quality estimation and evaluation of generated translations [Spe+20; Mat+20] by training models on *translation feedback* from humans, but this progress is, by and large, not yet reflected in either training or decoding methods. Exceptions such as minimum risk training [She+16; Edu+18] come at a cost of more expensive and unstable training, often with modest quality improvements.

An appealing alternative is to modify the decoding procedure only, separating it into two stages: *candidate generation* (where candidates are generated with beam search or sampled from the whole distribution) and *ranking* (where they are scored using a quality metric of interest, and the translation with the highest score is picked). This strategy has been explored in approaches using *N*-best reranking [Ng+19; Bha+21] and minimum

¹While this is effectively a *alignment* problem between the models training/decoding objective and translation quality, the reason for this misalignment aren't fully clear.

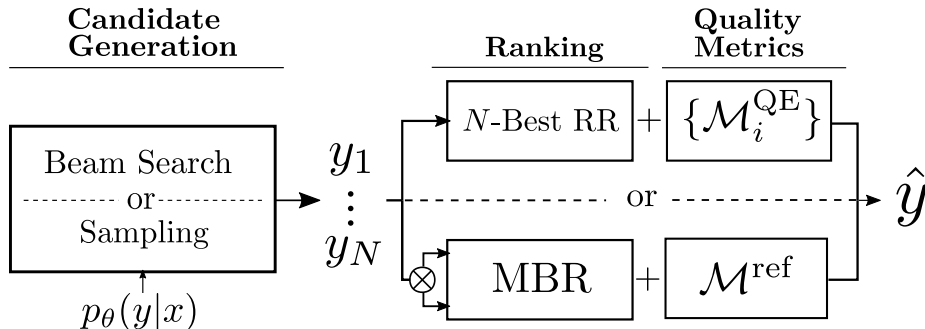


Figure 3.1: Quality-aware decoding framework. First, translation candidates are *generated* according to the model. Then, using reference-free and/or reference-based MT metrics, these candidates are *ranked*, and the highest ranked one is picked as the final translation.

Bayes risk (MBR) decoding [SN17; EA21; MS21]. While this previous work has exhibited promising results, it has mostly focused on optimizing lexical metrics such as BLEU or METEOR [Pap+02b; LD09], which have limited correlation with human judgments [MBC20; Fre+21a] and only loosely aligning with translation quality. Moreover, a rigorous apples-to-apples comparison among this suite of techniques and their variants is still missing, even though they share similar building blocks.

Our work fills these gaps by asking the question:

“Can we leverage recent advances in MT quality evaluation to generate better translations? If so, how can we most effectively do so?”

To answer this question, we systematically explore NMT decoding using a suite of ranking procedures. We take advantage of recent state-of-the-art learnable metrics, both reference-based, such as COMET and BLEURT [Rei+20a; SDP20], and reference-free (also known as *quality estimation*; QE), such as TransQuest and OpenKiwi [ROM20; Kep+19]. We compare different ranking strategies under a unified framework, which we name **quality-aware decoding**. First, we analyze the performance of decoding using N -best reranking, both *fixed* according to a single metric and *learned* using multiple metrics, where the coefficients for each metric are optimized according to a reference-based metric. Second, we explore ranking using reference-based metrics directly through MBR decoding. Finally, to circumvent the expensive computational cost of the latter when the number of candidates is large, we develop a two-stage ranking procedure, where we use N -best reranking to pick a subset of the candidates to be ranked through MBR decoding. We explore the interaction of these different ranking methods with various candidate generation procedures including beam search, vanilla sampling, and nucleus sampling.

Experiments with two model sizes and four datasets reveal that while MAP-based decoding appears competitive when evaluating with lexical-based metrics (BLEU and ChrF), the story is very different with state-of-the-art evaluation metrics, where quality-aware decoding shows significant gains, both with N -best reranking and MBR decoding. We perform a human-study to more faithfully evaluate our systems and find that, while performance on learnable metrics is not always predictive of the best system, quality-aware decoding usually results in translations with higher quality than MAP-based decoding.

3.2 Candidate Generation and Ranking

3.2.1 Candidate Generation

An NMT model defines a probability distribution $p_\theta(y|x)$ over a set of hypotheses $\mathcal{Y}$, conditioned on a source sentence x , where θ are learned parameters. A translation is typically predicted using MAP decoding, formalized as

$$\hat{y}_{\text{MAP}} = \operatorname{argmax}_{y \in \mathcal{Y}} \log p_\theta(y|x). \quad (3.1)$$

In words, MAP decoding searches for the most probable translation under $p_\theta(y|x)$, *i.e.*, the mode of the model distribution. Finding the exact $\hat{y}_{\text{MAP}}$ is intractable since the search space $\mathcal{Y}$ is combinatorially large, thus, approximations like **beam search** [Gra12a; SVL14] are used. However, it has been shown that the translation quality *degrades* for large values of the beam size [KK17; YHM18; MC18; MCV20], with the empty string often being the true MAP hypothesis [SB19].

A stochastic alternative to beam search is to *draw samples* directly from $p_\theta(y|x)$ with ancestral sampling, optionally with variants that truncate this distribution, such as top- k sampling [FLD18] or **p -nucleus sampling** [Hol+20] – the latter samples from the smallest set of words whose cumulative probability is larger than a predefined value p . Deterministic methods combining beam and nucleus search have also been proposed [SL21].

Unlike beam search, sampling is not a search algorithm nor a decision rule – it is not expected for a single sample to outperform MAP decoding [EA20]. However, samples from the model can still be useful for alternative decoding methods, as we shall see. While beam search focus on high probability candidates, typically similar to each other, sampling allows for more *exploration*, leading to higher candidate *diversity*.

3.2.2 Ranking

We assume access to a set $\bar{\mathcal{Y}} \subseteq \mathcal{Y}$ containing N candidate translations for a source sentence, obtained with one of the generation procedures described in §3.2.1. As long as N is relatively small, it is possible to (re-)rank these candidates in a post-hoc manner, such that the best translation maximizes a given metric of interest. We highlight two different lines of work for ranking in MT decoding: first, **N -best reranking**, using reference-free metrics as features; second, **MBR decoding**, using reference-based metrics.

N -best Reranking

In its simplest form (which we call *fixed* reranking), a *single* feature h is used (*e.g.*, an estimated quality score), and the candidate that maximizes this score is picked as the final translation,

$$\hat{y}_{\text{F-RR}} = \operatorname{argmax}_{y \in \bar{\mathcal{Y}}} h(x, y). \quad (3.2)$$

When *multiple* features $[h_1, \dots, h_K]$ are available, one can tune weights $[w_1, \dots, w_K]$ for these features to maximize a given reference-based evaluation metric on a validation set

[Och03; DK08] – we call this *tuned* reranking. In this case, the final translation is

$$\hat{y}_{\text{T-RR}} = \operatorname{argmax}_{y \in \bar{\mathcal{Y}}} \sum_{k=1}^K w_k h_k(x, y). \quad (3.3)$$

Minimum Bayes Risk (MBR) Decoding

While the techniques above rely on *reference-free* metrics for the computation of features, MBR decoding uses *reference-based* metrics to rank candidates. Unlike MAP decoding, which searches for the most probable translation, MBR decoding aims to find the translation that maximizes the expected *utility* (equivalently, that minimizes *risk*, [KB02; KB04; EA20]).² Let again $\bar{\mathcal{Y}} \subseteq \mathcal{Y}$ be a set containing N hypotheses and $h(x, y^*, y)$ a utility function measuring the similarity between a hypothesis $y \in \mathcal{Y}$ and a reference $y^* \in \bar{\mathcal{Y}}$, optionally conditioned on the source (*e.g.*, an automatic evaluation metric such as BLEU or COMET). MBR decoding seeks for

$$\begin{aligned} \hat{y}_{\text{MBR}} &= \operatorname{argmax}_{y \in \bar{\mathcal{Y}}} \underbrace{\mathbb{E}_{Y \sim p_{\theta}(y|x)}[h(x, Y, y)]}_{\approx \frac{1}{M} \sum_{j=1}^M h(x, y^{(j)}, y)}, \end{aligned} \quad (3.4)$$

where in Eq. 3.4 the expectation is approximated as a Monte Carlo (MC) sum using model samples $y^{(1)}, \dots, y^{(M)} \sim p_{\theta}(y|x)$ ³. In practice, the translation with the highest expected utility can be computed by comparing each hypothesis $y \in \bar{\mathcal{Y}}$ to all the other hypotheses in the set.

3.3 Quality-Aware Decoding

While recent works have explored various combinations of candidate generation and ranking procedures for NMT [LAR21; Bha+21; EA21; MS21], they suffer from two limitations:

- The ranking procedure is usually based on simple lexical-based metrics (BLEU, chrF, METEOR). Although these metrics are well established and inexpensive to compute, they correlate poorly with human judgments at segment level [Mat+20; Fre+21a].
- Each work independently explores N -best reranking or MBR decoding, making unclear which method produces better translations.

In this chapter, we hypothesize that using more powerful metrics in the ranking procedure may lead to better quality translations. We propose a unified framework for ranking with both reference-based and reference-free metrics, independently of the candidate generation procedure. We explore four methods with different computational costs for a given number of candidates, N .

²Utility can be seen as the translation quality of the hypothesis when compared to the reference.

³We also consider the case where $y^{(1)}, \dots, y^{(M)}$ are obtained from nucleus sampling or beam search. Although the original MC estimate is unbiased, these ones are biased.

Fixed N -best Reranker. An N -best reranker using a single reference-free metric as a feature, according to Eq. 3.2. The computational cost of this reranker is $\mathcal{O}(N \times C_{\mathcal{M}^{\text{QE}}})$, where $C_{\mathcal{M}^{\text{QE}}}$ denotes the cost of running an evaluation with a metric $\mathcal{M}^{\text{QE}}$.

Tuned N -best Reranker. An N -best reranker using as features *all* the reference-free metrics in, along with the model log-likelihood $\log p_{\theta}(y|x)$. The weights in Eq. 3.3 are optimized to maximize a given reference-based metric $\mathcal{M}^{\text{ref}}$ using MERT [Och03], a coordinate-ascent optimization algorithm widely used in previous work. Note that $\mathcal{M}^{\text{ref}}$ is used for tuning only; at test time, only reference-free metrics are used. Therefore, the decoding cost is $\mathcal{O}(N \times \sum_i C_{\mathcal{M}_i^{\text{QE}}})$.

MBR Decoding. Choosing as the utility function a reference-based metric $\mathcal{M}^{\text{ref}}$, we estimate the utility using a simple Monte Carlo sum, as shown in Eq. 3.4. The estimation requires computing pairwise comparisons and thus the cost of running MBR decoding is $\mathcal{O}(N^2 \times C_{\mathcal{M}^{\text{ref}}})$.

N -best Reranker $\rightarrow$ MBR. Using a large number of samples in MBR decoding is expensive due to its quadratic cost. To circumvent this issue, we explore a *two-stage* ranking approach: we first rank all the candidates using a tuned N -best reranker, followed by MBR decoding using the top M candidates. The computational cost becomes $\mathcal{O}(N \times \sum_i C_{\mathcal{M}_i} + M^2 \times C_{\mathcal{M}^{\text{ref}}})$. The first ranking stage *prunes* the candidate list to a smaller, higher quality subset, making possible a more accurate estimation of the utility with less samples, and potentially allowing a better reranker than *plain* MBR for almost the same computational budget.

3.3.1 Reference-based Metrics

Reference-based metrics are the standard way to evaluate MT systems; the most used ones rely on the lexical overlap between hypotheses and reference translations [Pap+02b; LD09; Pop15]. However, lexical-based approaches have important limitations: they have difficulties recognizing correct translations that are paraphrases of the reference(s); they ignore the source sentence, an important indicator of meaning for the translation; and they do not always correlate well with human judgments, particularly at segment-level [Fre+22]

In this work, apart from BLEU and chrF, we use the following state-of-the-art trainable reference-based metrics for both ranking and performance evaluation of MT systems:

- BLEURT [SDP20; Pu+21], trained to regress on human direct assessments (DA; [Gra+13]). We use the largest multilingual version, *BLEURT-20*, based on the RemBERT model [Chu+21].
- COMET [Rei+20a], based on XLM-R [Con+20], trained to regress on quality assessments such as DA using both the reference and the source to assess the quality of a given translation. We use the publicly available model developed for the WMT20 metrics shared task (*wmt20-comet-da*).

These metrics have shown much better correlation at segment-level than previous lexical metrics in WMT metrics shared tasks [Mat+20; Fre+21c], mainly because they are directly trained to approximate human translation feedback (§2.2.3). Hence, as discussed in §3.2.2, they are good candidates to be used either *indirectly* as an optimization objective for learning the tuned reranker’s feature weights, or *directly* as a utility function in MBR decoding. In the former, the higher the metric correlation with human judgment, the better the translation picked by the tuned reranker. In the latter, we approximate the expected utility in Eq. 3.4 by letting a candidate generated by the model be a reference translation – a suitable premise *if* the model is good in expectation.

3.3.2 Reference-free Metrics

MT evaluation metrics have also been developed for the case where references are not available – they are called *reference-free* or *quality estimation* (QE) metrics. These more closely approximate how humans experts give feedback to translations [Fre+21a], and more closely approximate feedback models trained for other tasks, but have generally lacked when compared to reference-based metrics due the extra difficulty of the task. However in the last years, considerable improvements have been made to such metrics, with state-of-the-art models having increasing correlations with human annotators [Fre+21c; Spe+21]. These improvements enable the use of such models for ranking translation hypotheses in a more reliable way than before.

In this chapter, we explore four recently proposed reference-free metrics as features for N -best reranking, all at the sentence-level:

- COMET-QE [Rei+20b], a reference-free version of COMET (§3.3.1). It was the winning submission for the QE-as-a-metric subtask of the WMT20 shared task [Mat+20].
- TransQuest [ROM20], the winning submission for the sentence-level DA prediction subtask of the WMT20 QE shared task [Spe+20]. Similarly to COMET-QE this metric predicts a DA score.
- MBART-QE [Zer+21a], based on the mBART [Liu+20] model, trained to predict both the *mean* and the *variance* of DA scores. It was a top performer in the WMT21 QE shared task [Spe+21].
- OpenKiwi-MQM [Kep+19; Rei+21], based on XLM-R, trained to predict the *multi-dimensional quality metric* (MQM; [LUB14]).⁴ This reference-free metric was ranked second on the QE-as-a-metric subtask from the WMT 2021 metrics shared task.

3.4 Experiments

3.4.1 Setup

We study the benefits of quality-aware decoding over MAP-based decoding in two regimes:

⁴MQM annotations are expert-level type of annotations more fine-grained than DA, with individual errors annotated. See Table 2.1 for an example.

- A high-resource, unconstrained, setting with *large* transformer models (6 layers, 16 attention heads, 1024 embedding dimensions, and 8192 hidden dimensions) trained by Ng et al. [Ng+19] for the WMT19 news translation task [Bar+19], using English to German (EN $\rightarrow$ DE) and English to Russian (EN $\rightarrow$ RU) language pairs. These models were trained on over 20 million parallel and 100 million back-translated sentences, being the winning submissions of that year’s shared task. We consider the non-ensembled version of the model and use *newstest19* for validation and *newstest20* for testing.
- A more constrained scenario with a *small* transformer model (6 layers, 4 attention heads, 512 embedding dimensions, and 1024 hidden dimensions) trained from scratch in *Fairseq* [Ott+19] on the smaller IWSLT17 datasets [CGF12] for English to German (EN $\rightarrow$ DE) and English to French (EN $\rightarrow$ FR), each with a little over 200k training examples. We chose these datasets because they have been extensively used in previous work [Bha+21] and smaller model allows us to answer questions about how the training methodology affects ranking performance (see § ??).

We use beam search with a beam size of 5 as our decoding baseline because we found that it resulted in better or similar translations than larger beam sizes. For tuned N -best reranking, we use Travatar’s [Neu13] implementation of MERT [Och03] to optimize the weight of each feature, as described in §3.3.2. Finally, we evaluate each system using the metrics discussed in §3.3.1, along with BLEU and chrF [Pop15].

3.4.2 Results

Overall, given all the metrics, candidate generation, and ranking procedures, we evaluate over 150 systems per dataset. We report subsets of this data separately to answer specific research questions, and defer to Appendix ?? for additional results.

Impact of Candidate Generation

First, we explore the impact of the candidate generation procedure and the number of candidates.

Which candidate generation method works best, beam search or sampling?

We generate candidates with beam search, vanilla sampling, and nucleus sampling. For the latter, we use $p = 0.6$ based on early results showing improved performance for all metrics.⁵ For N -best reranking, we use up to 200 samples; for MBR decoding, due to the quadratic computational cost, we use up to 100.

Figure 3.2 shows BLEU and COMET for different candidate generation and ranking methods for the EN $\rightarrow$ DE WMT20 and IWSLT17 datasets, with increasing number of candidates. The baseline is represented by the dashed line. To assess the performance *ceiling* of the rankers, we also report results with an *oracle* ranker for the reported metrics,

⁵We picked nucleus sampling over top- k sampling because it allows varying support size and has outperformed top- k in text generation tasks [Hol+20].

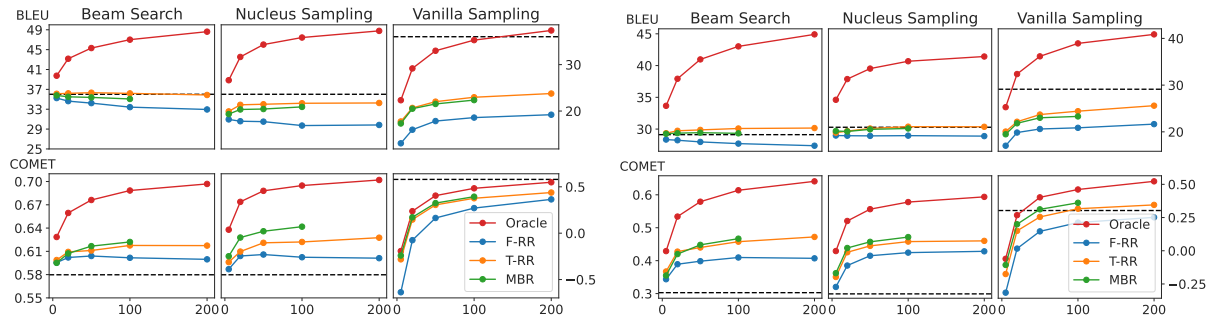


Figure 3.2: Values for BLEU (top) and COMET (bottom) for EN $\rightarrow$ DE as we increase the number of candidates for different generation and ranking procedures, as well as oracles with the respective metrics, for the *large* (left) and *small* (right) models. Baseline values (with beam size of 5) are marked with a dashed horizontal line.

picking the candidate that maximizes it. For the *fixed* N -best reranker, we use COMET-QE as a metric, albeit the results for other reference-free metrics are similar. Performance seems to scale well with the number of candidates, particularly for vanilla sampling and for the *tuned* N -best reranker and MBR decoder. [LAR21; MS21]. However, all the rankers using vanilla sampling severely under-perform the baseline in most cases (see also §??). In contrast, the rankers using beam search or nucleus sampling are competitive or outperform the baseline in terms of BLEU, and greatly outperform it in terms of COMET. For the larger models, we see that the performance according to the lexical metrics degrades with more candidates. In this scenario, rankers using nucleus sampling seem to have an edge over the ones that use beam search for COMET.

Based on the findings above, and due to generally better performance of COMET over BLEU for MT evaluation [Koc+21a], in following experiments we use nucleus sampling with the *large* model and beam search with the *small* model.

Impact of Ranking and Metrics

We now investigate the usefulness of the metrics presented in §3.3 as features and objectives for ranking. For N -best reranking, we use all the available candidates (200) while, for MBR, due to the computational cost of using 100 candidates, we report results with 50 candidates only (we found that ranking with *tuned* N -best reranking with $N = 100$ and MBR with $N = 50$ takes about the same time). We report results in Table 3.1, and use them to answer some specific research questions.

Which QE metric works best in a fixed N -best reranker? We consider a *fixed* N -best reranker with a single reference-free metric as a feature (see Table 3.1, second group). While none of the metrics allows for improving the baseline results in terms of the lexical metrics (BLEU and chrF), rerankers using COMET-QE or MBART-QE outperform the baseline according to BLEURT and COMET, for both the *large* and *small* models. Due to the aforementioned better performance of these metrics for translation quality evaluation,

	Large (WMT20)				Small (IWSLT)			
	BLEU	chrF	BLEURT	COMET	BLEU	chrF	BLEURT	COMET
Baseline	36.01	63.88	0.7376	0.5795	29.12	56.23	0.6635	0.3028
F-RR w/ COMET-QE	29.83	59.91	<u>0.7457</u>	0.6012	<u>27.38</u>	54.89	<u>0.6848</u>	<u>0.4071</u>
F-RR w/ MBART-QE	<u>32.92</u>	<u>62.71</u>	0.7384	0.5831	27.30	<u>55.62</u>	0.6765	0.3533
F-RR w/ OpenKiwi	30.38	59.56	0.7401	0.5623	25.35	51.53	0.6524	0.2200
F-RR w/ Transquest	31.28	60.94	0.7368	0.5739	26.90	54.46	0.6613	0.2999
T-RR w/ BLEU	<u>35.34</u>	<u>63.82</u>	0.7407	0.5891	30.51	57.73	0.7077	0.4536
T-RR w/ BLEURT	33.39	62.56	<u>0.7552</u>	0.6217	30.16	57.40	<u>0.7127</u>	<u>0.4741</u>
T-RR w/ COMET	34.26	63.31	0.7546	<u>0.6276</u>	30.16	57.32	0.7124	0.4721
MBR w/ BLEU	<u>34.94</u>	<u>63.21</u>	0.7333	0.5680	29.25	56.36	0.6619	0.3017
MBR w/ BLEURT	32.90	62.34	<u>0.7649</u>	0.6047	28.69	56.28	<u>0.7051</u>	0.3799
MBR w/ COMET	33.04	62.65	0.7477	<u>0.6359</u>	<u>29.43</u>	<u>56.74</u>	0.6882	<u>0.4480</u>
T-RR+MBR w/ BLEU	<u>35.84</u>	63.96	0.7395	0.5888	<u>30.23</u>	<u>57.34</u>	0.6913	0.3969
T-RR+MBR w/ BLEURT	33.61	62.95	0.7658	0.6165	29.28	56.77	0.7225	0.4361
T-RR+MBR w/ COMET	34.20	63.35	0.7526	0.6418	29.46	57.13	0.7058	0.5005

Table 3.1: Evaluation metrics for EN $\rightarrow$ DE for the *large* and *small* model settings, using a *fixed* N -best reranker (F-RR), a *tuned* N -best reranker (T-RR), MBR decoding, and a two-stage approach. Best overall values are **bolded** and best for each specific group are underlined.

we hypothesize that these rankers produce better translations than the baseline. However, since the sharp drop in the lexical metrics is concerning, we will verify this hypothesis in a human study, in §3.4.2.

How does the performance of a tuned N -best reranker vary when we change the optimization objective? We consider a *tuned* N -best reranker using as features *all* the reference-free metrics in §3.3.2, and optimized using MERT. Table 3.1 (3rd group) shows results for EN $\rightarrow$ DE. For the *small* model, all the rankers show improved results over the baseline for all the metrics. In particular, optimizing for BLEU leads to the best results in the lexical metrics, while optimizing for BLEURT leads to the best performance in the others. Finally, optimizing for COMET leads to similar performance than optimizing for BLEURT. For the *large* model, although none of the rerankers is able to outperform the baseline in the lexical metrics, we see similar trends as before for BLEURT and COMET.

How does the performance of MBR decoding vary when we change the utility function? Table 3.1 (4th group) shows the impact of the utility function (BLEU, BLEURT, or COMET). For the *small* model, using COMET leads to the best performance according to all the metrics except BLEURT (for which the best result is attained when optimizing itself). For the *large* model, the best result according to a given metric is obtained when using that metric as the utility function.

How do (tuned) N -best reranking and MBR compare to each other? Looking at Table 3.1 we see that, for the *small* model, N -best reranking seems to perform better than MBR decoding in all the evaluation metrics, including the one used as the utility

	EN-DE (WMT20)					EN-RU (WMT20)				
	BLEU	chrF	BLEURT	COMET	Human R.	BLEU	chrF	BLEURT	COMET	Human R.
Reference	-	-	-	-	4.51	-	-	-	-	4.07
Baseline	36.01	63.88	0.7376	0.5795	4.28	23.86	51.16	0.6953	0.5361	3.62
F-RR w/ COMET-QE	29.83	59.91	0.7457	0.6012	4.19	20.32	49.18	0.7130	0.6207	3.25
T-RR w/ COMET	34.26	63.31	0.7546	0.6276	4.33	22.42	50.91	0.7243	0.6441	3.65
MBR w/ COMET	33.04	62.65	0.7477	0.6359	4.27	23.67	51.18	0.7093	0.6242	3.66
T-RR + MBR w/ COMET	34.20	63.35	0.7526	0.6418	4.30	23.21	51.26	0.7238	0.6736	3.72[†]

	EN-DE (IWSLT17)					EN-FR (IWSLT17)				
	BLEU	chrF	BLEURT	COMET	Human R.	BLEU	chrF	BLEURT	COMET	Human R.
Reference	-	-	-	-	4.38	-	-	-	-	4.00
Baseline	29.12	0.6635	56.23	0.3028	3.68	38.12	0.6532	63.20	0.4809	3.92
F-RR w/ COMET-QE	27.38	0.6848	54.89	0.4071	3.67	35.59	0.6628	60.90	0.5553	3.63
T-RR w/ COMET	30.16	0.7124	57.32	0.4721	3.90[†]	38.60	0.7020	63.77	0.6392	4.05 [†]
MBR w/ COMET	29.43	0.6882	56.74	0.4480	3.79 [†]	37.77	0.6710	63.24	0.6127	4.05 [†]
T-RR + MBR w/ COMET	29.46	0.7058	57.13	0.5005	3.83 [†]	38.33	0.6883	63.53	0.6610	4.09[†]

Table 3.2: Results for automatic and human evaluation. Top: WMT20 (large models); Bottom: IWSLT17 (small models). Methods with [†] are statistically significantly better than the baseline, with $p < 0.05$.

function in MBR decoding. The picture is less clear for the *large* model, with MBR decoding achieving best values for a given fine-tuned metric when using it as the utility; this comes at the cost of worse performance according to the other metrics, hinting at a potential “*overfitting*” effect. Overall, N -best reranking seems to have an edge over MBR decoding. We will further clarify this question with human evaluation in § 3.4.2.

Can we improve performance by combining N -best reranking with MBR decoding? Table 3.1 shows that, for both the *large* and the *small* model, the two-stage ranking approach described in §3.3 leads to the best performance according to the fine-tuned metrics. In particular, the best result is obtained when the utility function is the same as the evaluation metric. These results suggest that a promising research direction is to seek more sophisticated pruning strategies for MBR decoding.

Human Evaluation

Which metric correlates more with human judgments? How risky is it to optimize a metric and evaluate on a related metric? Our experiments suggest that, overall, *quality-aware* decoding produces translations with better performance across most metrics than *MAP-based* decoding. However, for some cases (such as fixed N -best reranking and most results with the *large* model), there is a concerning “metric gap” between lexical-based and fine-tuned metrics. While the latter have shown to correlate better with human judgments, previous work has not attempted to explicitly optimize these metrics, and doing so could lead to ranking systems that learn to exploit “pathologies” in these metrics rather than improving translation quality. To investigate this hypothesis, we perform a human study across all four datasets. We ask annotators to rate, from 1 (no overlap in meaning) to 5 (perfect translation), the translations produced by the 4 *ranking* systems in §3.3, as well as the baseline translation and the reference. We choose COMET-QE as the feature for the fixed N -best ranker and COMET as the optimization metric

and utility function for the tuned N -best reranker and MBR decoding, respectively. The reasons for this are two-fold: (1) they are currently the reference-free and reference-based metrics with highest reported correlation with human judgments [Koc+21a], (2) we saw the largest “metric gap” for systems based on these metrics, hinting of a potential “overfitting” problem (specially since COMET-QE and COMET are similar models).

Table 3.2 shows the results for the human evaluation, as well as the automatic metrics. We see that, with the exception of T-RR w/ COMET, when fine-tuned metrics are explicitly optimized for, their correlation with human judgments decreases and they are no longer reliable indicators of system-level ranking. This is notable for the fixed N -best reranker with COMET-QE, which outperforms the baseline in COMET for every single scenario, but leads to markedly lower quality translations. However, despite the potential for overfitting these metrics, we find that *tuned* N -best reranking, MBR, and their combination consistently achieve better translation quality than the baseline, specially with the small model. In particular, N -best reranking results in better translations than MBR, and their combination is the best system in 2 of 4 LPs.

3.5 Related Work

Reranking. Inspired by the work of Shen et al. [SSO04] on discriminative reranking for SMT, Lee et al. [LAR21] trained a large transformer model using a reranking objective to optimize BLEU. Our work differs in which our rerankers are much simpler and therefore can be tuned on a validation set; and we use more powerful quality metrics instead of BLEU. Similarly, Bhattacharyya et al. [Bha+21] learned an energy-based reranker to assign lower energy to the samples with higher BLEU scores. While the energy model plays a similar role to a QE system, our work differs in two ways: we use an existing, pretrained QE model instead of training a dedicated reranker, making our approach applicable to any MT system without further training; and the QE model is trained to predict human assessments, rather than BLEU scores. Leblond et al. [Leb+21] compare a reinforcement learning approach to reranking approaches (but not MBR decoding, as we do). They investigate the use of reference-based metrics and, for the reward function, a reference-free metric based on a modified BERTScore [Zha+20]. This new multilingual BERTScore is not fine-tuned on human judgments as COMET and BLEURT and it is unclear what its level of agreement with human judgments is. Another line of work is *generative reranking*, where the reranker is not trained to optimize a metric, but rather as a generative noisy-channel model [Yu+17; YDA19; Ng+19].

Minimum Bayes Risk Decoding. MBR decoding [KB02; KB04] has recently been revived for NMT using candidates generated with beam search [Sta+17; SN17] and sampling [EA20; MS21]. Eikema and Aziz [EA21] also explore a two-stage approach for MBR decoding. Additionally, there is concurrent work by Freitag et al. [Fre+21b] on using neural metrics as utility functions during MBR decoding: however they limit their scope to MBR with reference-based metrics, while we perform a more extensive evaluation over ranking methods and metrics. Amrhein and Sennrich [AS22] also concurrently explored using MBR

decoding with neural metrics, but with the purposes of identifying weaknesses in the metric (in their case COMET), similarly to the *metric overfitting* problem we discussed in §3.4.2. A comparison with N -best re-ranking was missing in these works, a gap this chapter fills. A related line of work is *minimum risk training* (MRT; [SE06; She+16]), which *trains* models to minimize risk, allowing arbitrary non-differentiable loss functions [Edu+18; Wie+19] and avoiding exposure bias [WS20; KK21]. However, MRT is considerably more expensive and difficult to train and the gains are often small. Incorporating our quality metrics in MRT is an exciting research direction.

3.6 Conclusions & Subsequent Work

We leverage recent advances in MT quality estimation and evaluation and propose *quality-aware decoding* for NMT, effectively aligning MT with translation quality at decoding-time. We explore different candidate generation and ranking methods, with a comprehensive empirical analysis across four datasets and two model classes. We show that, compared to MAP-based decoding, quality-aware decoding leads to better translations, according to powerful automatic evaluation metrics and human judgments.

Since the publication of this work [Fer+22b], the use of neural metrics as part of the MT pipeline has become popular technique to increase the final quality of translations. Follow-up works have studied how to improve the efficiency of MBR to make a more practical approach [CV23], but neural metrics have also been used as feedback models to directly optimise the NMT parameters, particularly when using LLMs as MT models [Gul+23; Ram+23].

Fine-grained MT Evaluation with Large Language Models

In the previous chapter (Chapter 3), we showed that we can leverage feedback models of translation quality to align NMT models at decoding time. However, these feedback models require large amounts of human feedback and might suffer from problems of *reward overoptimization* (§3.4.2). In this chapter, we show that multilingual LLMs can be used to provide *AI feedback* (§2.2.4) for MT, comparable to state-of-the-art learned metrics and propose **AutoMQM**, a prompting technique that allows LLMs to provide more fine-grained and interpretable feedback of translation quality.

The outputs of our models prompted with AUTOMQM are available at github.com/google-research/google-research.

4.1 Introduction

Evaluating natural language generation systems has always been challenging, and as the output quality of these systems has improved, evaluation has become even more challenging and critical. For example, in Machine Translation (MT), a field where evaluation has garnered considerable attention, previous standard automatic surface-level metrics such as BLEU [Pap+02b] are becoming less reliable as the quality of generation systems improves, and *learned* automatic metrics that leverage human-judgments to finetune language models (like the ones explored in previous chapter) currently represent the state-of-the-art in automatic evaluation benchmarks like the WMT Metrics task [Fre+22], and show high correlation with human judgments. However, these metrics typically output a single, *uninterpretable* quality score, making it difficult to understand the type and extent of errors identified by them. The lack of insights makes it difficult for model developers to leverage these metrics to improve their systems.

Unlike automatic metrics that only provide a single scalar value as quality score, state-of-the-art human evaluation methodologies like Multidimensional Quality Metrics [MQM; LUB14; Fre+21a] ask professional annotators to identify and label error spans with a

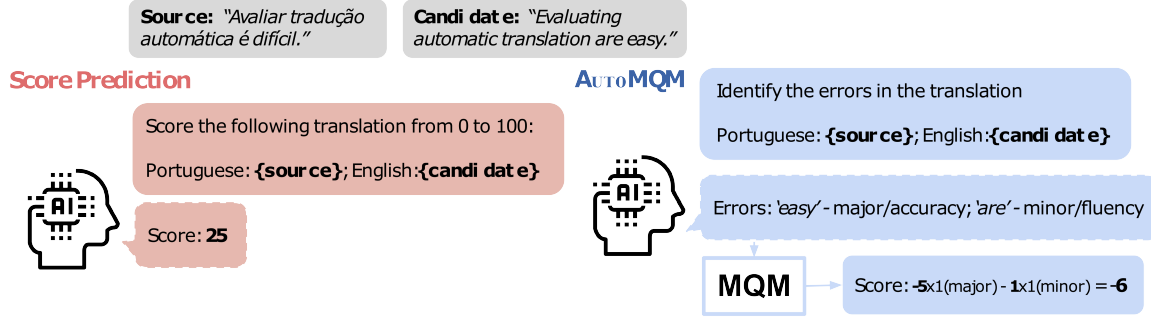


Figure 4.1: Illustration of how AUTOMQM uses LLMs to assess the quality of a translation. Rather than asking for a single quality score, AUTOMQM prompts models to identify and classify errors, and uses the MQM framework to produce a score.

category and severity. This much richer feedback can be used to gain a better understanding of the current limitations of the model under evaluation and improve it.

In this chapter, we ask whether large language models (LLMs) in combination with a few human annotations can be used to design an automatic metric that generates rich feedback similar to that generated by human experts in MQM. This chapter is motivated by recent papers that demonstrated that LLMs can be used as automatic metrics [Liu+23b] to generate a single quality score. In particular, Kocmi and Federmann [KF23] showed that LLMs can be prompted to assess the quality of machine-generated translations, even achieving state-of-the-art performance on assessing system-level quality. However, previous work only provides a limited view of the capabilities of LLMs for machine translation evaluation: the focus has predominantly been on *score prediction* (i.e. predicting a numerical value for quality), without considering the use of *any* annotated data (either through in-context learning or finetuning), and only in *high-resource* language pairs.

We provide a large-scale study of the capabilities of LLMs [from the PaLM and PaLM-2 families; Cho+22; Ani+23] for machine translation evaluation (both with and without a reference translation), provide a novel comparison between prompting and finetuning. Inspired by findings that the performance of LLMs can be improved by prompting them for *rationales* of their predictions [Wei+22; Lu+23], we also propose **AUTOMQM**, a prompting technique for MT evaluation that asks LLMs to identify error spans in a translation and to classify these errors according to the MQM framework, with a quality score derived automatically from the identified errors. A key advantage of AUTOMQM is its *interpretability*, as users can inspect the errors responsible for a score (Figure 4.1).

Our contributions can be summarized as follows:

- We confirm the finding of [KF23] that LLMs are *zero-shot* state-of-the-art system-level evaluators, but show low correlation with human judgment compared to *learned* metrics at the segment-level.
- We show that *finetuning* an LLM with human judgment mitigates its low segment-level performance (particularly for smaller LLMs), showing similar correlations with human judgment at both the system-level and segment-level to state-of-the-art learned

metrics.

- We find that, with AUTOMQM, PaLM-2 models can be prompted to generate rich MQM-like annotations, outperforming their score prediction counterparts at the segment-level.
- Furthermore, annotations predicted by PaLM-2 models correctly identify over 50% of words that are part of *major* errors, and are comparable to the ones produced by state-of-the-art *supervised* word-level evaluators.

Our findings might have significant implications for not only MT evaluation, but evaluation of machine-generated text in general, and further highlight the potential of using LLMs to provide *AI Feedback* [Fer+23a].

4.2 Background: MT Evaluation

Machine translation evaluation is one of the most well-studied evaluation problems in NLP [Cal+08; Fre+22]. In this task, given

1. a *source* sentence in a (source) language
2. a *candidate* translation in a (target) language

an evaluation metric assesses the quality of the candidate translation by how well it conveys the meaning of the source sentence while considering other factors like *fluency*. Like many other natural language generation evaluation problems, this task is difficult because the set of correct translations for a given source sentence is often very large and not entirely known in advance. To simplify the problem of machine translation evaluation, often (3) a *reference* translation (typically created by a professional human translator) is included as additional information when assessing the candidate translation. This sub-problem is known as *reference-based* evaluation (as opposed *reference-less* evaluation or *quality estimation*).

Up until recently, human evaluation of machine translation was carried out predominantly with the aim of assigning a single quality score to a candidate translation. Consequently, *learned* metrics, which leverage collected human judgment data, are trained for and evaluated on the same task of *score prediction* (i.e., assigning a single quality score to a candidate translation), and can achieve high correlation with human-provided scores [Fre+22]. However, framing machine translation evaluation as a score prediction task is problematic: any scoring or ranking of translations is implicitly based on an identification of errors in the candidate translations, and asking raters to solely provide a single score can lead to rushed and noisy judgments [Fre+21a].

This insight has led to the adoption of the Multidimensional Quality Metrics (MQM) framework [LUB14; Fre+21a] as the gold standard for evaluating machine translation. The MQM framework asks human evaluators to identify error spans in candidate translations and classify those errors according to various dimensions, e.g., *fluency*, *accuracy*, ... Importantly, the MQM framework *does not* ask annotators to provide a quality score for each translation, and instead derives one automatically from the identified error spans and their classifications. However, despite its richness, *most* automatic metrics that leverage MQM

Based on the given source and reference, identify the major and minor errors in this translation. Note that Major errors refer to actual translation or grammatical errors, and Minor errors refer to smaller imperfections, and purely subjective opinions about the translation.

```
{src_lang} source: "{source}"
{tgt_lang} human reference: "{reference}"
{tgt_lang} translation: "{candidate}"
Score (0-100): {score}
```

Figure 4.2: The *score prediction* prompt used in this chapter. (GEMBA-SQM [KF23]). Parts in purple are only included for *reference-based* evaluation, while parts in orange represent slots for outputs and are only included for in-context examples.

data only use the final quality score produced by the framework and discard the error span information and classification.

4.3 Using LLMs to Predict Quality Scores

Recent works have shown that large language models are versatile, general-purpose models that can be used to tackle many problems in NLP, including evaluation [KF23; Jai+23b; Liu+23b]. We begin by exploring how LLMs can be used for machine translation evaluation through *score prediction*.

4.3.1 Prompting

We start by measuring how far we can push the performance of LLMs with just *prompting* [Liu+23a]: by defining the task of MT evaluation and quality estimation as *textual templates* (with a general description of the problem and “slots” for the inputs and outputs), we can use general-purpose LLMs to perform these tasks at inference-time, without any parameter updates.

Throughout the chapter, we choose to use Kocmi and Federmann [KF23]’s GEMBA-SQM prompt (Figure 4.2), which asks models to generate (a string representation of) a score from 0-100. We choose this prompt for two reasons: firstly, early explorations with various prompts showed that this generally performed well. Secondly, using a single prompt ensures a fairer comparison between the capabilities of different models.

In-Context Learning A surprising emergent capability of LLMs is their ability to improve on prompting-based tasks by including a very small amount of labeled data as part of the prompt/context [Bro+20] and *without* parameter updates, a technique called *in-context learning* (ICL) or *few-shot prompting*. We thus investigate the impact that ICL has on LLMs’ ability to assess translation quality. Recent works have shown that the impact of ICL is tightly tied with the exact examples included in the prompt, with a poor selection procedure leading to no improvements or even worse performance than the zero-shot case [Jai+23b]. We therefore explore two sampling approaches to select in-context

examples from a pre-defined “pool” of translation quality assessments: **uniform** and **stratified sampling**, where the example pool is bucketed by score ranges and examples are sampled from each bucket.

4.3.2 Finetuning

It has previously been shown that LLMs are capable of zero-shot evaluation [KF23], but the extent to which *finetuning* on human judgment data can further boost the performance of LLMs has not been studied. In the WMT’22 Metrics Shared Task [Fre+22], all top submissions were learned metrics; that is, pretrained models finetuned on human feedback.¹

Thus, we investigate whether LLMs are amenable to finetuning on human judgment data. LLMs used in top-performing metrics are generally much larger than the pretrained language models leveraged by previous learned metrics (which generally have fewer than 1 billion parameters). Moreover, most learned metrics leverage pretrained encoder-only rather than (decoder-only) prefix language models. We experiment with finetuning LLMs using two objectives:

- *Regression (R)*: Commonly used for training learned metrics [Rei+22a], the objective here is a regression loss (e.g., mean squared error) between continuous scores obtained from the model (for example, with a *regression head*) and the human scores.
- *Generative Classification (GC)*: We bucket scores into discrete classes (e.g. “*bad*”, “*ok*” and “*good*”) and treat the MT evaluation task as a text-to-text classification problem [Raf+20b] by having the model generate a template sentence with the class.

4.4 Using LLMs to Predict Error Spans

While producing quality scores that correlate with human judgments is an important part of translation quality assessment, metrics that solely do score prediction suffer from problems of **interpretability**: if a metric assigns a low score, the downstream users are left in the dark about which parts of the translation were responsible for the score and thus need to be corrected. This is especially problematic in cases where the metric assigns a *wrong* score to a translation, as it is much harder to diagnose why the evaluation model made a mistake, and identify and prevent similar mistakes in the future. In fact, reducing translation quality to a single score has proven problematic even for human annotators: asking raters to solely provide a single score can lead to rushed and noisy judgments [Fre+21a] and the current gold standard for translation quality evaluation involving human annotators is instead based on methodologies like the MQM framework (see §5.2), which provide richer feedback by identifying error spans, categorizing them, and evaluating their severity.

Interestingly, another emergent phenomenon in LLMs is the success of *chain-of-thought* prompting [Wei+22]: when defining a prompt for a particular task, if we instruct the model to produce a series of intermediate reasoning steps (“*let’s think step-by-step*”), it tends to

¹While these metrics all leverage powerful pretrained (language) models, these generally aren’t considered LLMs

generate a free-text *rationale* before generating an output, and this often improves the performance on the task at hand [Liu+23b]. Furthermore, this *chain-of-thought* prompting can be used to obtain *structured* rationales from LLMs, and this can lead to better performance than with free-text rationales [Lu+23].

Motivated by these findings, we propose **AUTO MQM**, a prompting technique for translation quality assessment that instructs LLMs to *identify* errors in a translation, and *categorize* the type of error according to the MQM framework [LUB14]. Furthermore, we *don't* ask the model to produce a score, as the MQM framework provides an algorithmic procedure to obtain one from identified errors: the total score is the sum of penalties for all errors identified, where (roughly) *major* errors get penalized with -5 and *minors* with -1 .² Figure 4.3 shows the main AUTO MQM prompt used in this chapter.

```
Based on the given source and reference, identify the major and minor errors in this translation. Note that Major errors refer to actual translation or grammatical errors, and Minor errors refer to smaller imperfections, and purely subjective opinions about the translation.

{src_lang} source: "{source}"
{tgt_lang} human reference: "{reference}"
{tgt_lang} translation: "{candidate}"
Errors: {error1:span} - {error1:severity}/{error1:category}; {error2:span} - ...
```

Figure 4.3: The AUTO MQM prompt used in this chapter. Parts in purple are only included for *reference-based* evaluation, while parts in orange represent slots for outputs, and are only included for in-context examples.

Importantly, obtaining meaningful AUTO MQM results in a zero-shot setting is a substantially more challenging task compared to score prediction: we found that, without any in-context examples, LLMs tend to produce outputs that are either uninformative or difficult to parse. Thus we only consider the AUTO MQM task in the *few-shot* scenario. Based on the findings from §4.5.2, we explore the impact of in-context learning by sampling from the example pool using stratified sampling extended with a set of *rejection criteria*, which ensures that the example set has a balance between major and minor errors as well as diversity in the categories of errors.

4.5 Experiments

4.5.1 Experimental Setup

Data The metrics in this work are evaluated on three *high-resource* high-resource language pairs come from the WMT’22 Metrics Shared Task [Fre+22]: en→de, zh→en, and en→ru. The ground-truth translation quality scores are derived from MQM ratings in which expert annotators marked error spans in the translations with different severity levels which are automatically converted to a numeric score (see §5.2). See Table 4.1 for statistics about the number of MT systems and segments for each language pair.

²This is similar to methods that leverage external *executors* to improve LLMs [Gao+22].

Additionally, in our experiments, AUTOMQM required in-context examples with MQM annotations to work, so we restrict our evaluation of AUTOMQM to en→de and zh→en because there are available MQM ratings from the WMT’21 Metrics Shared Task [Fre+21c] that we can use as in-context learning example pools.

Models We base most of our experiments on the following LLMs:

- **PaLM:** A 540 billion parameter autoregressive Transformer model trained on 780 billion tokens of high-quality text [Cho+22]. It showed remarkable performance on a wide-range of NLP tasks, including Machine Translation [Vil+22].
- **PaLM-2:** The successor to PaLM, the PaLM-2 family of LLMs [Ani+23] builds upon recent research insights, such as compute-optimal scaling, a more multilingual and diverse pre-training mixture, and architectural/optimization improvements. We mainly use two model sizes in the family: PaLM-2 BISON and (the larger) PaLM-2-UNICORN.³ In addition we explore the impact of instruction-tuning by using a UNICORN model finetuned on the FLAN dataset [Wei+21].

We compare to a set of strong baseline evaluation metrics, MetricX-XXL and COMET-22, which were the two top-performing metrics in the WMT’22 Metrics Shared Task. MetricX-XXL and COMET-22 are both finetuned regression models trained on DA data from WMT that are initialized with mT5 [Xue+21] and XLM-R [Con+20], respectively.

For the AUTOMQM experiments, we also compare against MATESE, a comparable submission to the WMT’22 Metrics Shared task that finetuned an XLM-R model to identify major and minor errors, and computed a score automatically. Since we were unable to obtain the span-level predictions for the MATESE submission, we also compare against the top submission to the WMT’22 Word-Level Quality Estimation Shared Task [Zer+21b]: word-level COMETKIWI (COMET-WL) [Rei+22b], also based on an XLM-R model trained on a combination of sentence- and word-level data. To do so, we re-run this model on the WMT’22 Metrics Shared Task data, and convert the predicted *word-level* OK/BAD tags into spans.⁴

Finetuning For *regression* finetuning, we use a real-valued logit, extracted from a fixed index in the first target token’s logit vector, as the quality signal. (In particular, we leverage a special, *unused*, vocabulary token.) This was the technique used to train MetricX-XXL in

³Information about exact number of parameters of PaLM-2 models is not publicly available.

⁴We consider a span as any maximal consecutive sequence of words marked as BAD, assigning every span the *major* severity.

LP	#Sys	#Seg
en→de	13	1315
zh→en	14	1875
en→ru	15	1315

Table 4.1: The number of systems and segments that have MQM scores, used as ground-truth human feedback.

the WMT 2022 Shared Task submission [Fre+22]. The regression-based model was trained on WMT direct assessment (DA) data from the years 2015 through 2020.

For *generative* classification, we bucket the scores in the training data into five classes, where class boundaries are assigned so that each class contains an equal number of training examples. We then map labels to verbal ratings from the following set, based on their bucket: [”very bad”, ”bad”, ”ok”, ”good”, ”very good”]. To evaluate the model, predictions are mapped back to integer labels from 1 to 5. Any predictions not containing a substring in the label set are considered invalid and are mapped to 0. We experimented with finetuning on both DA and MQM 2020 [Fre+21a] data, and found that the latter performed slightly better.

To assess the impact of *model size*, we also finetune two additional (smaller) PaLM-2 models, which we call *S* and *M*, comparing their finetuned and zero-shot performance.⁵

Metric Meta-Evaluation The quality of an automatic evaluation metric is estimated by comparing the agreement between the metric scores and ground-truth quality scores on a large number of translations from different MT systems, a process known as metric meta-evaluation. This work reports three different agreement scores, as follows.

The first is system-level accuracy, which calculates the percent of system pairs that are ranked the same by the metric and ground-truth scores, micro-averaged over a set of language pairs [Koc+21b]. System-level scores are defined as the average score across all segments.

At the segment-level, the standard correlation that is reported by WMT is Kendall’s τ . However, recent work pointed out problems with Kendall’s τ with respect to ties [DFF23]. In short, different variants of τ are inconsistent with respect to ties and even biased against metrics that predict ties, as our metrics do in this work. Deutsch et al. [DFF23] recommend reporting a pairwise accuracy score, which rewards metrics for correctly ranking translations as well as correctly predicting ties, in combination with a tie calibration procedure that automatically introduces ties into metric scores so that the meta-evaluation is fairer. This accuracy score, denoted acc^* , ranges between 0 and 1, and a random metric would achieve 33% accuracy. We report the “group-by-item” variant of the pairwise accuracy score from Deutsch et al. [DFF23] in addition to Pearson’s ρ , a complementary signal to rank-based correlations that measure the strength of the linear relationship between two variables (and one of the standard correlations reported in WMT).

Span Meta-Evaluation Since AUTOMQM provides not only scores but also the identified error spans, we can compare the predicted spans with the errors marked by annotators in the MQM annotations. We evaluate quality of predicted spans using: (1) *Span Precision* (SP), which measures the overlap of predicted spans and gold (annotated) spans; and (2) *Major recall* (MR), which captures the percentage of gold major errors that were predicted as errors (either minor or major).

⁵We use a small variation of the *zero-shot* prompt, asking models for scores from the same 5 buckets used in finetuning.

More formally, consider the set of ground truth spans S^* , where each span consists of a sequence of words, i.e., $s_i = (w_{(a)}, w_{(a+1)}, \dots)$. Let $S_{\text{maj}}^* \subseteq S^*$ be the subset containing only the major errors. Given a span set S , we define its positional set $P(S)$ as the set containing the positions of all the words in every span in S . For example, assuming a span $s_i = (w_{(n)}, w_{(n+1)}, \dots)$ in S starts at the n th position in the text, its corresponding positional set will include the positions $\{n, n+1, \dots, n + \text{len}(s_i) - 1\}$. Then for a set of *predicted* spans $\hat{S}$, SP and MR are defined as:

$$\text{SP}(\hat{S}) = \frac{|P(\hat{S}) \cap P(S^*)|}{|P(\hat{S})|} \quad (4.1)$$

$$\text{MR}(\hat{S}) = \frac{|P(\hat{S}) \cap P(S_{\text{maj}}^*)|}{|P(S_{\text{maj}}^*)|} \quad (4.2)$$

Intuitively, we care for overall precision (regardless of severity) since we want to make sure predicted errors tend to be marked by annotators as well, but for recall we care mostly for *major* errors, as these have a larger impact on translation quality and are more critical to identify. Additionally, we also report the (3) *Matthews Correlation Coefficient* (MCC), one of the official metrics in the word-level quality estimation tasks [Zer+22].

4.5.2 Results

Model	Ref?	System-Level			Segment-Level			
		All (3 LPs)	EN-DE		ZH-EN		EN-RU	
		Accuracy	ρ	acc*	ρ	acc*	ρ	acc*
Baselines								
MetricX-XXL	✓	85.0%	0.549	61.1%	0.581	54.6%	0.495	60.6%
COMET-22	✓	83.9%	0.512	60.2%	0.585	54.1%	0.469	57.7%
COMET-QE	✗	78.1%	0.419	56.3%	0.505	48.8%	0.439	53.4%
Prompting								
PaLM 540B	✓	90.1%	0.247	55.4%	0.255	48.5%	0.180	48.6%
PaLM-2 BISON	✓	88.7%	0.394	56.8%	0.322	49.3%	0.322	52.8%
PaLM-2 UNICORN	✓	90.1%	0.401	56.3%	0.349	51.1%	0.352	55.3%
FLAN-PaLM-2 UNICORN	✓	75.9%	0.197	55.6%	0.139	46.1%	0.198	52.0%
PaLM 540B	✗	84.3%	0.239	56.1%	0.270	43.1%	0.300	51.8%
PaLM-2 BISON	✗	85.0%	0.355	57.0%	0.299	48.6%	0.303	53.1%
PaLM-2 UNICORN	✗	84.3%	0.275	56.1%	0.252	48.3%	0.209	49.8%
FLAN-PaLM-2 UNICORN	✗	69.7%	0.116	54.6%	0.112	43.8%	0.156	47.8%
Finetune								
PaLM-2 BISON (R)	✓	88.0%	0.511	61.0%	0.459	51.5%	0.458	59.5%
PaLM-2 BISON (GC)	✓	86.1%	0.400	59.2%	0.444	49.3%	0.365	56.0%
PaLM-2 UNICORN (R)	✓	87.6%	0.508	61.1%	0.412	52.6%	0.460	60.4%
PaLM 2 BISON (R)	✗	87.6%	0.490	59.9%	0.439	53.4%	0.437	59.2%
PaLM 2 BISON (GC)	✗	86.1%	0.368	57.5%	0.420	47.3%	0.390	54.9%
PaLM 2 UNICORN (GC)	✗	86.1%	0.407	57.9%	0.402	45.6%	0.411	55.3%

Table 4.2: Meta-evaluation results at system and segment-level for the *high-resource* language pairs. Finetuned (**R**) and (**GC**) represent the *regression* and *generative classification* objectives (§4.3.2). ✓ and ✗ represent *reference-based* and *reference-less* metrics, respectively.

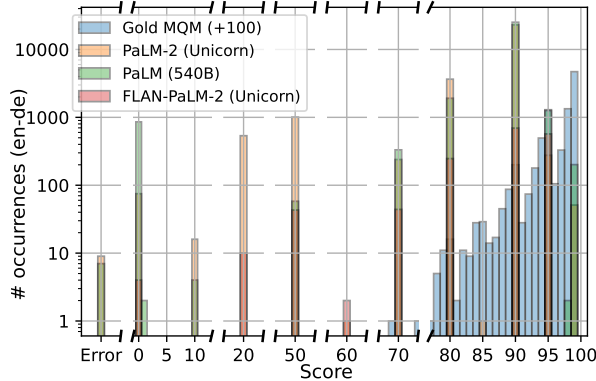


Figure 4.4: Distribution of scores for various LLM *reference-based* evaluators, on the EN-DE test set. Note that the y axis is in *log-scale*.

Score Prediction

Table 4.2 summarizes the meta-evaluation results, at the *system* and *segment* level, for both the *zero-shot prompting* and *finetuning* settings.

Prompting A first observation is almost all zero-shot LLM evaluators have higher *system-level* performance than learned metrics (with and without references), with PaLM 540B and PaLM-2 UNICORN achieving the best performance. At the segment level, the story is more complicated: similarly to Kocmi et al. [Koc+22], we find that none of the LLMs we explored was able to consistently outperform the baseline learned metrics. We see that PaLM-540B is a particularly poor reference-based evaluator, which is surprising given its system-level performance. Unexpectedly, instruction-tuning with FLAN seems to *degrade* performance, with FLAN-PaLM-2 UNICORN achieving poor performance at both the system and segment levels.⁶

Nevertheless, PaLM-2 models achieve high correlations with human judgments, and the *reference-less* PaLM-2 BISON is competitive with the *learned* baselines, particularly at assessing alternative translations of the same sentence (acc^*).

Figure 4.4 shows the distribution of scores produced by PaLM- and PaLM-2-based evaluators. We find that, despite being prompted to give a score in the 0-100 range, these models almost always output one of a very limited set of scores (e.g. 0, 50, 90, 95). Given Kocmi and Federmann [KF23]’s similar findings with GPT models, it seems that this is a consequence of the pretraining objective.

Finetuning Despite their already-great performance in the zero-shot setting, we find that finetuning LLMs can further improve LLM evaluators’ segment-level scores. This is particularly obvious for the *reference-less* evaluators, where a finetuned PaLM-2 BISON achieves state-of-the-art performance in segment-level correlations and comparable system-level accuracy across all language pairs. Moreover, when we look at how performance *scales*

⁶Note that this might be a problem with the FLAN dataset and not instruction-tuning in general, as the GPT models are also instruction-tuned and perform well.

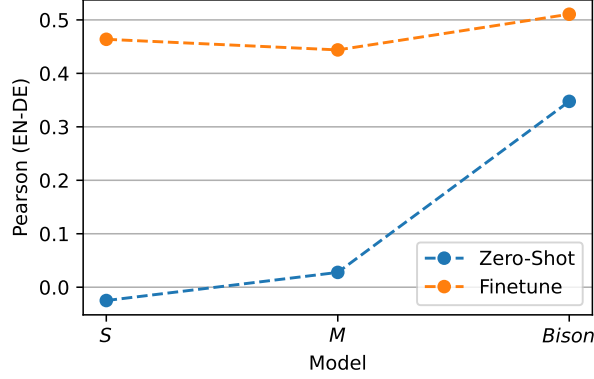


Figure 4.5: Behavior of *Pearson* as we scale the LLM’s parameter count. Note that the x axis is not to-scale with regard to parameter count.

with parameter count (Figure 4.5), we observe an interesting trend: while smaller models are not capable of being effective zero-shot evaluators, finetuning them leads to competitive performance, and only a slight decrease when compared to their larger finetuned counterparts.

In-context Learning Figure 4.6 shows the mean and interquartile range (IQR) of the performance as we increase the number of in-context examples k (with 100 example sets per k) sampled with *stratified* sampling. Surprisingly, despite evidence of the benefits of in-context learning for many tasks, we found that including in-context examples during evaluation (almost) never led to better performance, either with *uniform* or *stratified* sampling.

AutoMQM

Figure 4.7 shows the mean and interquartile range (IQR) of the performance of PaLM-2 BISON with AUTOMQM, as we increase the number of in-context examples (again, with

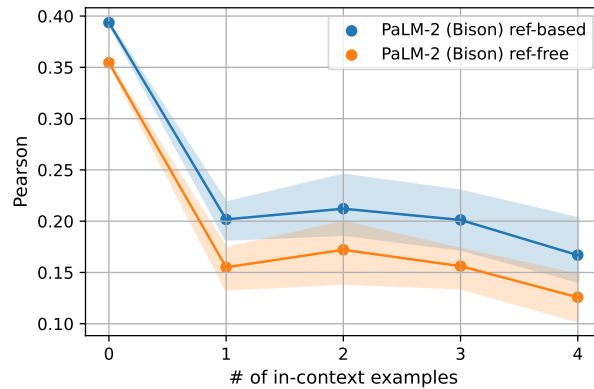


Figure 4.6: Mean *Pearson* and its interquartile range (IQR) in the WMT22 EN-DE test set, as we increase the number of in-context examples with *stratified* sampling

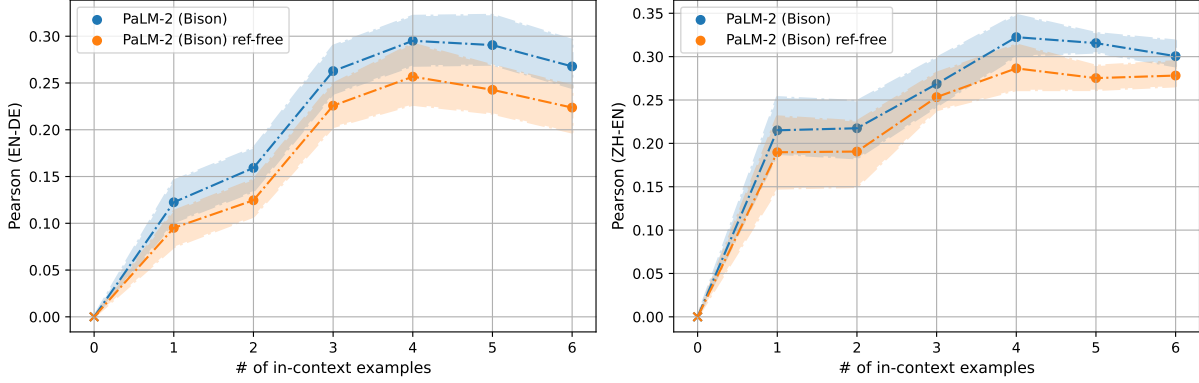


Figure 4.7: Mean *Pearson* and its interquartile range (IQR), as we increase the number of in-context examples in the AUTOMQM prompt, for EN-DE (left) and ZH-EN (right).

100 example sets per k). Contrary to the performance with score prediction, we find that performance with AUTOMQM seems to (mostly) scale with the number of in-context examples: performance increases monotonically with up to 4 in-context examples and plateaus thereafter. Additionally, the variance across the in-context learning sets seems to be lower, with most example sets exhibiting less than 0.05 *Pearson* difference from the best-performing sets. All this suggests that LLM evaluators are much more robust to the choice of in-context examples when prompted for AUTOMQM rather than for score prediction. We also find that the behavior of in-context learning is quite similar for both reference-based and reference-less evaluation tasks.

Model	Ref?	System-Level	Segment-Level			
		All (2 LPs)	EN-DE		ZH-EN	
		Accuracy	ρ	acc*	ρ	acc*
Baselines						
MetricX-XXL	✓	81.1%	0.549	61.1%	0.581	54.6%
MATESE	✓	79.9%	0.391	58.8%	0.528	51.5%
COMET-QE	✗	76.9%	0.419	56.3%	0.505	48.8%
MATESE-QE	✗	73.4%	0.298	57.9%	0.468	50.1%
COMET-WL	✗	71.6%	0.418	57.1%	0.406	51.5%
Score Prediction						
PaLM-2 BISON	✓	86.4%	0.394	56.8%	0.322	49.3%
PaLM-2 UNICORN	✓	86.4%	0.401	56.3%	0.349	51.1%
PaLM-2 BISON	✗	84.0%	0.355	57.0%	0.299	48.6%
PaLM-2 UNICORN	✗	80.5%	0.275	56.1%	0.252	48.3%
AutoMQM						
PaLM-2 BISON	✓	84.0%	0.369	59.2%	0.355	48.4%
PaLM-2 UNICORN	✓	87.6%	0.432	59.1%	0.442	51.8%
PaLM 2 BISON	✗	87.6%	0.297	55.2%	0.331	48.0%
PaLM 2 UNICORN	✗	83.4%	0.368	56.4%	0.429	50.2%

Table 4.3: Meta-evaluation results for PaLM-2 models using *AutoMQM* and score prediction, at the system and segment levels for multiple language pairs.

Table 4.3 shows the meta-evaluation results for PaLM-2 BISON and UNICORN prompted with AUTOMQM (using the best-performing in-context learning sets in Figure 4.7). For

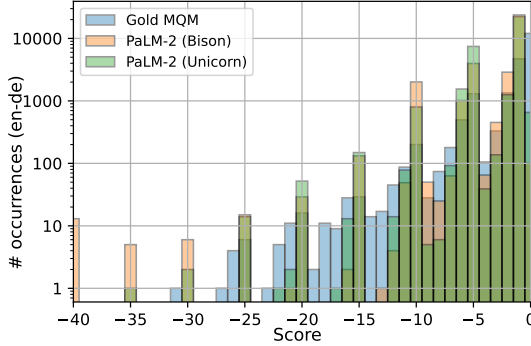


Figure 4.8: Distribution of scores for PaLM-2 models using AUTOMQM, on WMT22 EN-DE

ease of comparison, we also report their performance when prompted for *score prediction*, as well as the performance of the baselines. Overall, prompting LLMs with AUTOMQM seems to lead to significant improvements in evaluating machine translation quality, particularly for larger models: UNICORN achieves better performance (across all meta evaluations) with it than when prompted for *score prediction*, and its reference-less version is competitive with the best learned metric even at the segment level. However, for the smaller BISON, the benefits of AUTOMQM are less clear, with both techniques performing comparably. This hints that *scale* is necessary for *zero-* and *few-* shot fine-grained evaluation (like with AUTOMQM). We also find that the *distribution* of scores produced by LLMs prompted with AUTOMQM is much closer to the gold MQM distribution, with models outputting a much larger set of scores, and in the same ranges as annotators do (see Figure 4.8).

Model	R?	EN-DE			ZH-EN		
		SP	MR	MCC	SP	MR	MCC
Baselines							
COMET-WL	✗	0.267	0.250	0.161	0.364	0.178	0.152
AutoMQM							
BISON	✓	0.095	0.749	0.060	0.252	0.255	0.109
UNICORN	✓	0.175	0.628	0.193	0.238	0.476	0.143
BISON	✗	0.119	0.520	0.092	0.224	0.311	0.091
UNICORN	✗	0.150	0.580	0.150	0.229	0.488	0.133

Table 4.4: Span-level meta-evaluation on WMT22 for PaLM-2 models using *AutoMQM*. **SR** and **MR** represent *span precision* and *major recall*, respectively.

Finally, when evaluating the error spans produced by LLMs prompted with AUTOMQM (Table 4.4), we find that PaLM-2 models are able to identify most of the *major* errors. However, it does seem to *over-predict* errors (with errors predicted by UNICORN having on average ~ 5 words per span vs ~ 2 words in the ground truth) and have overall low span precision. Similarly to overall *score* correlations, *scale* also seems to be important for the

quality of spans produced by AUTOMQM, with UNICORN outperforming BISON at most metrics. Additionally, UNICORN prompted with AutoMQM predicts spans of comparable quality to the ones produced by current state-of-the-art *learned* word-level evaluators (trained on a considerable number of fine-grained annotations derived from MQM): while word-level models are more precise, their overall span correlation (MCC) is comparable, and they miss considerably more *major* errors than LLMs (despite only leveraging a handful of annotations).

4.6 Related Work

The success of *learned* machine translation metrics [SDP20; Rei+22a; Fre+22; Qin+22], which finetune neural network models pretrained on large amounts of (unsupervised) data, highlighted the importance of leveraging *transfer learning* to achieve metrics with better correlation with human judgments. More recently, *generative* LLMs [Ope23; Ani+23] have consistently demonstrated impressive results in natural language understanding and *zero*- and *few-shot* transfer and, naturally, interest in employing these models for (translation) evaluation has increased. Kocmi and Federmann [KF23] first explored the use of GPT models for evaluating machine translation tasks, showing their potential as *zero-shot* evaluators, and others have since extended GPT-based evaluation to other generation problems [Jai+23b; Liu+23b].

Perrella et al. [Per+22] first highlighted that MQM annotations could be leveraged to allow pretrained models to predict major and minor errors and, similarly to AUTOMQM, used the identified errors to automatically score translations. However, their approach relied on weaker encoder-only or encoder-decoder language models, required *supervised* data to work, and overall underperformed other top metrics. We compare against their *MaTASe* metric in our experiments. Lu et al. [Lu+23] showed that doing *error analysis*, a prompting technique similar to AUTOMQM, could lead to better ChatGPT-based evaluators. However, they still relied on the LLM to provide a score once it identified errors (rather than do it automatically using something like the MQM framework). Furthermore, they provided a very limited meta-evaluation using only 40 examples per language pair. Concurrently with our work, Xu et al. [Xu+23b] proposed INSTRUCTSCORE, a LLaMA-based evaluator that asks models to identify and categorize errors in translation (as well as providing a natural language explanation for each error). However, the authors only explore a 7B parameter model and don't leverage zero- and few-shot capabilities of models as in this work. Instead, they rely on a more complex approach of distilling the knowledge of a more capable GPT-4 LLM.

Additionally, WMT Word-Level Quality Estimation shared tasks [Fon+19; Zer+22] leverage MQM data by converting span-level annotations of errors (normally of *major* severity) to word-level tags and Task 2 in the WMT19 Quality Estimation shared task evaluation explicitly evaluated submissions of span-level annotations (although most submissions still consisted of models that predicted word-level tags which were converted to spans). We also compare against state-of-the-art word-level quality estimation models.

4.7 Conclusion & Subsequent Work

In this study, we have systematically investigated the capabilities of large language models for machine translation evaluation through *score prediction*, and proposed AUTOMQM, a novel prompting technique that leverages the Multidimensional Quality Metrics (MQM) framework for interpretable MT evaluation using LLMs. We demonstrated that just prompting LLMs for score prediction leads to state-of-the-art system-level evaluators, but still falls short of the best *learned* metrics at the segment-level (with finetuning being necessary to close this gap). Then we showed that AUTOMQM can further improve the performance of LLMs without finetuning while providing interpretability through error spans that align with human annotations.

Our findings surrounding finetuning LLMs for *score prediction* hint that LLMs’ performance in machine translation evaluation could be further improved by finetuning these models on fine-grained human judgment data (like MQM). In fact, subsequent work has since shown that training evaluation models on fine-grained MQM can lead to state-of-the-art metrics that surpass ones trained only on sentence-level scores [Gue+23], and that fine-grained feedback produced by these models can be used to improve the quality of translations produced by LLMs [Xu+23c].

Part II

Model Explanations as Learning Signals

Optimizing Model Explanations for Teaching

In the previous part, we explored how models of human feedback can be used to align translation models (Chapter 3) and how LLMs can be used to provide such feedback with minimal human supervision (Chapter 4). In this chapter, we now move to the problem of interpretability: by assuming explanations as having similar goals to feedback (§2.3.2), we propose **SMaT**, a novel method for extracting explanations from a trained (teacher) neural network model, showing that these explanations help other student models *simulate* the teacher and that these explanations correlate with human explanations of the same decision (without ever being optimized for it).

The code for the experiments in this chapter is available open-source at github.com/coderpat/learning-scaffold.

5.1 Introduction

While deep learning’s performance has led it to become the dominant paradigm in machine learning, its relative opaqueness has brought great interest in methods to improve *model interpretability*. Many recent works propose methods for extracting *explanations* from neural networks (section 5.6), which vary from the highlighting of relevant input features [SVZ14; Arr+17; DXK19] to more complex representations of the reasoning of the network [MA20; Wu+21]. However, are these methods actually achieving their goal of making models more interpretable? Some concerning findings have cast doubt on this proposition; different explanations methods have been found to disagree on the same model/input [Nee+21; Bas+21] and explanations do not necessarily help predict a model’s output and/or its failures [Cha+18].

In fact, the research community is still in the process of understanding *what* explanations are supposed to achieve, and *how* to assess success of an explanation method [DK17; Mil19]. Many early works on model interpretability designed their methods around a set of desiderata [STY17b; LT19] and relied on qualitative assessment of a handful of samples with respect to these desiderata; a process that is highly subjective and is hard to reproduce. In contrast, recent works have focused on more quantitative criteria: correlation

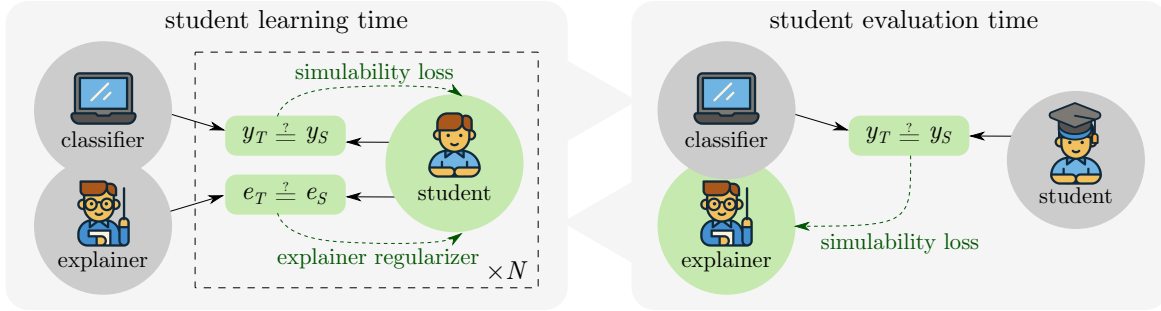


Figure 5.1: Illustration of our SMaT framework. First, a student model is trained to recover the classifier’s predictions and to match the explanations given by the explainer. Then, the explainer is updated based on how well the trained student *simulates* the classifier (without access to explanations). In practice, we repeat these two consecutive processes for several steps. Green arrows and boxes represent learnable components.

between explainability methods for measuring *consistency* [JW19; SS19], *sufficiency* and *comprehensiveness* [DeY+20], and *simulability*: whether a human or machine consumer of explanations understands the model behavior well enough to predict its output on unseen examples [Lip16a; DK17]. Simulability, in particular, has a number of desirable properties, such as being intuitively aligned with the goal of *communicating* the underlying model behavior to humans and being measurable in manual and automated experiments [TM20; HB20; Pru+20a].

For instance, Pruthi et al. [Pru+20a] proposed a framework for automatic evaluation of simulability that, given a *teacher model* and explanations of this model’s predictions, trains a *student model* to match the teacher’s predictions. The explanations are then evaluated with respect to how well they help a student *learn to simulate* the teacher (section 5.2). This is analogous to the concept in pedagogy of **instructional scaffolding** [VVB10], a process through which a teacher adds support for students to aid learning. More effective scaffolding—in our case, better explanations—is assumed to lead to better student learning. However, while this previous work provides an attractive way to *evaluate* existing explanation methods, it stops short of proposing a method to actually *improve* them.

In this chapter, we propose to *learn to explain* by directly learning explanations that provide better scaffolding of the student’s learning, a framework we term **Scaffold-Maximizing Training (SMaT)**. Figure 5.1 illustrates the framework: the explainer is used to *scaffold* the student training, and is updated based on how well the student does at *test* time at simulating the teacher model. We take insights from research on meta-learning [FAL17; Rag+21], formalizing our setting as a bi-level optimization problem and optimizing it based on higher-order differentiation (section 5.3). Importantly, our high-level framework makes few assumptions about the model we are trying to explain, the structure of the explanations or the modalities considered. To test our framework, we then introduce a *parameterized* attention-based explainer optimizable with SMaT that works for any model with attention mechanisms (section 5.4).

We experiment with SMaT in text classification, image classification, and (multilin-

gual) text-based regression tasks using pretrained transformer models (section 5.5). We find that our framework is able to effectively optimize explainers across all the considered tasks, where students trained with *learned* attention explanations achieve better simulability than baselines trained with *static* attention or gradient-based explanations. We further evaluate the *plausability* of our explanations (i.e., whether produced explanations align with how people would justify a similar choice) using human-labeled explanations (text classification and text regression) and through a human study (image classification) and find that explanations learned with SMaT are more plausible than the static explainers considered. Overall, the results reinforce the utility of scaffolding as a criterion for evaluating and improving model explanations.

5.2 Evaluating Explanations with Simulability

Consider a model $T : \mathcal{X} \rightarrow \mathcal{Y}$ trained on some dataset $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$. For example, this could be a text or image classifier that was trained on a particular downstream task (with $\mathcal{D}_{\text{train}}$ being the training data for that task). *Post-hoc* interpretability methods typically introduce an *explainer* module $E_T : \mathcal{T} \times \mathcal{X} \rightarrow \mathcal{E}$ that takes a model and an input, and produces an explanation $e \in \mathcal{E}$ for the output of the model given that input, where $\mathcal{E}$ denotes the space of possible explanations. For instance, interpretability methods using saliency maps define $\mathcal{E}$ as the space of *normalized* distributions of importance over L input elements $e \in \Delta_{L-1}$ (where Δ_{L-1} is the $(L-1)$ -probability simplex).

[Pru+20a] proposed an automatic framework for evaluating explainers that trains a *student* model $S_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ with parameters θ to *simulate* the *teacher* (i.e., the original classifier) in a *constrained* setting. For example, the student can be constrained to have less capacity than the teacher by using a simpler model or trained with a subset of the dataset used for the teacher ($\hat{\mathcal{D}}_{\text{train}} \subsetneq \mathcal{D}_{\text{train}}$).

In this framework, a baseline student S_θ is trained according to

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} [\mathcal{L}_{\text{sim}}(S_\theta(x), T(x))]$$

and its simulability $\text{SIM}(S_{\theta^*}, T)$ is measured on an unseen test set. The actual form of $\mathcal{L}_{\text{sim}}$ and $\text{SIM}(S_{\theta^*}, T)$ is task-specific. For example, in a classification task, we use cross-entropy as the simulation loss $\mathcal{L}_{\text{sim}}$ over the teacher’s predictions, while the simulability of a model S_{θ^*} can be defined as the simulation accuracy, i.e., what percentage of the student and teacher predictions match over a *held-out* test set $\mathcal{D}_{\text{test}}$:

$$\text{SIM}(S_{\theta^*}, T) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}} [\mathbb{1}\{S_{\theta^*}(x) = T(x)\}]. \quad (5.1)$$

Next, the training of the student is augmented with explanations produced by the explainer E . We introduce a student explainer $E_S : \mathcal{S} \times \mathcal{X} \rightarrow \mathcal{E}$, (the *S-explainer*) to extract explanations from the student, and *regularizing* these explanations on the explanations of teacher (the *T-explainer*), using a loss $\mathcal{L}_{\text{expl}}$ that takes explanations for both models:

$$\theta_E^* = \underset{\theta}{\operatorname{argmin}} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} \left[\underbrace{\mathcal{L}_{\text{sim}}(S_\theta(x), T(x))}_{\text{simulability loss}} + \beta \underbrace{\mathcal{L}_{\text{expl}}(E_S(S_\theta, x), E_T(T, x))}_{\text{explainer regularizer}} \right]. \quad (5.2)$$

For example, Pruthi et al. [Pru+20a] considered as a teacher explainer E_T various methods such as LIME [RSG16], Integrated Gradients [STY17b], and attention mechanisms, and explored both attention regularization (using Kullback-Leibler divergence) and multi-task learning to regularize the student.

The key assumption surrounding this evaluation framework is that a student trained with *good* explanations should learn to simulate the teacher better than a student trained with bad or no explanations, that is, $\text{SIM}(S_{\theta_E^*}, T) > \text{SIM}(S_{\theta^*}, T)$. For clarity, we will refer to the simulability of a model $S_{\theta_E^*}$ trained using explanations as *scaffolded* simulability.

5.3 Optimizing Explainers for Teaching

As a **first contribution** of this chapter, we extend the previously described framework to make it possible to directly optimize the teacher explainer so that it can most effectively teach the student the original model’s behavior. To this end, consider a *parameterized* T -explainer E_{ϕ_T} with parameters ϕ_T , and equivalently a *parameterized* S -explainer E_{ϕ_S} with parameters ϕ_S . We can write the loss function for the student and S -explainer as:

$$\mathcal{L}_{\text{student}}(S_{\theta}, E_{\phi_S}, T, E_{\phi_T}, x) = \mathcal{L}_{\text{sim}}(S_{\theta}(x), T(x)) + \beta \mathcal{L}_{\text{expl}}(E_{\phi_S}(S_{\theta}, x), E_{\phi_T}(T, x)). \quad (5.3)$$

While this framework is flexible enough to rigorously and automatically evaluate many types of explanations, calculating scaffolded simulability requires an optimization procedure to learn the student and S -explainer parameters θ, ϕ_S . This makes it non-trivial to achieve our goal of directly finding the teacher explainer parameters ϕ_T that optimize scaffolded simulability. To overcome this challenge, we draw inspiration from the extensive literature on meta-learning [Sch87; FAL17], and frame the optimization as the following bi-level optimization problem (see Grefenstette et al. [Gre+19] for a primer):

$$\theta^*(\phi_T), \phi_S^*(\phi_T) = \underset{\theta, \phi_S}{\text{argmin}} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} [\mathcal{L}_{\text{student}}(S_{\theta}, E_{\phi_S}, T, E_{\phi_T}, x)] \quad (5.4)$$

$$\phi_T^* = \underset{\phi_T}{\text{argmin}} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}} [\mathcal{L}_{\text{sim}}(S_{\theta^*(\phi_T)}(x), T(x))]. \quad (5.5)$$

Here, the *inner* optimization updates the student and the S -explainer parameters (Equation 5.4), and in the *outer* optimization we update the T -explainer parameters (Equation 5.5). **Importantly**, our framework does not modify the teacher, as our goal is to explain a model without changing its original behavior. Notice that we also simplify the problem by considering the more tractable simulation loss $\mathcal{L}_{\text{sim}}$ instead of the simulability metric $\text{SIM}(S_{\theta^*}, T)$ as part of the objective for the outer optimization.

Now, if we assume the explainers E_{ϕ_T} and E_{ϕ_S} are differentiable, we can use gradient-based optimization [FAL17] to optimize both the student (with its explainer) and the T -explainer. In particular, we use *explicit* differentiation to solve this optimization problem. To compute gradients for ϕ_T , we have to differentiate through a gradient operation, which requires Hessian-vector products, an operation supported by most modern deep learning frameworks [Bra+18; Gre+19]. However, explicitly computing gradients for ϕ_T through a large number of inner optimization steps is computationally intractable. To circumvent

this problem, typically the inner optimization is run for only a couple of steps or a *truncated* gradient is computed [Sha+19]. In this chapter, we take the approach of taking a *single* inner optimization step and learning the student and S -explainer jointly with the T -explainer *without* resetting the student [Der+21]. At each step, we update the student and S -explainer parameters as follows:

$$\theta^{t+1} = \theta^t - \eta_{\text{INN}} \nabla_{\theta} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} \left[\mathcal{L}_{\text{student}}(S_{\theta^t}, E_{\phi_S^t}, T, E_{\phi_T^t}, x) \right] \quad (5.6)$$

$$\phi_S^{t+1} = \phi_S^t - \eta_{\text{INN}} \nabla_{\phi_S} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} \left[\mathcal{L}_{\text{student}}(S_{\theta^t}, E_{\phi_S^t}, T, E_{\phi_T^t}, x) \right]. \quad (5.7)$$

After updating the student, we take an extra gradient step with the new parameters but only use these updates to calculate the *outer*-gradient for ϕ_T , without actually updating θ . This approach is similar to the *pilot update* proposed by Zhou et al. [ZXM21], and we verified that it led to more stable optimization in practice:

$$\theta(\phi_T^t) = \theta^{t+1} - \eta_{\text{INN}} \nabla_{\theta} \mathbb{E}_{(x,y) \sim \hat{\mathcal{D}}_{\text{train}}} \left[\mathcal{L}_{\text{student}}(S_{\theta^{t+1}}, E_{\phi_S^{t+1}}, T, E_{\phi_T^t}, x) \right] \quad (5.8)$$

$$\phi_T^{t+1} = \phi_T^t - \eta_{\text{OUT}} \nabla_{\phi_T} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}} \left[\mathcal{L}_{\text{sim}} \left(S_{\theta(\phi_T^t)}(x), T(x) \right) \right]. \quad (5.9)$$

5.4 Parameterized Attention Explainer

As a **second contribution** of this chapter, we introduce a novel *parameterized* attention-based explainer that can be learned with our framework. Transformer models [Vas+17b] are currently the most successful deep-learning architecture across a variety of tasks [Sho+19; Wor+22]. Underpinning their success is the *multi-head attention mechanism*, which computes a *normalized* distribution over the $1 \leq i \leq L$ input elements in parallel for each head h :

$$A^h = \text{SOFTMAX}(Q^h(K^h)^\top), \quad (5.10)$$

where $Q^h = [q_0^h, \dots, q_L^h]$ and $K^h = [k_0^h, \dots, k_L^h]$ are the *query* and *key* linear projections over the input element representations for head h . Attention mechanisms have been used extensively for producing saliency maps [WP19; Vas+19] and while some concerns have been raised regarding their faithfulness [JW19], overall attention-based explainers have been found to lead to relatively good explanations in terms of *plausibility* and *simulability* [TM20; Kob+20; Pru+20a].

However, to extract explanations from multi-head attention, we have two important design choices:

1. **Single distribution selection:** Since self-attention produces an attention matrix $A^h \in \Delta_{L-1}^L$, we need to *pool* these attention distributions to produce a single saliency map $e \in \Delta_{L-1}$. Typically, the distribution from a single token (such as [CLS]) or the *average* of the attention distributions from all tokens $1 \leq i \leq L$ are used.
2. **Head selection:** We also need to *pool* the distributions produced by each head. Typical ad-hoc strategies include using the mean over all heads for a certain layer [FSA21] or selecting a single head based on plausibility on validation set [Tre+21].

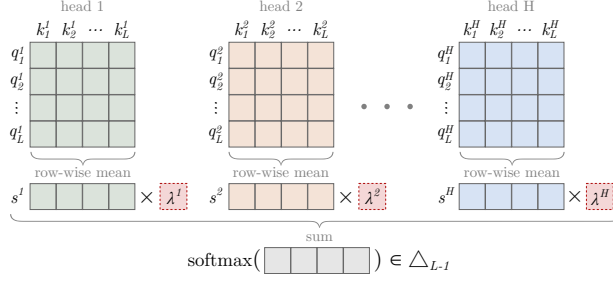


Figure 5.2: Our parameterized attention-based explainer. Dashed red boxes represent learned parameters $\lambda_T = \text{SPARSEMAX}(\phi_T) \in \Delta_{H-1}$, weighting average attention logits of each head $1 \leq h \leq H$. A softmax over the weighted sum generates the attention probabilities.

However, since transformers can have hundreds or even thousands of heads, these choices rely on human intuition or require large amounts of plausibility labels.

In this chapter, we approach the latter design choice in a more principled manner. Concretely, we associate each head with a weight and then perform a weighted sum over all heads. These weights are learned such that the resulting explanation maximizes simulability, as described in [section 5.3](#). More formally, given a model T_{θ_T} and its query and key projections for an input x for each layer and head $h \leq H$, we define a *parameterized, differentiable* attention explainer $E_{\phi_T}(T_{\theta_T}, x)$ as

$$s^h = \frac{1}{L} \sum_{i=1}^L (q_i^h)^\top K^h, \quad E_{\phi_T}(T, x) = \text{SOFTMAX} \left(\sum_{h=1}^H \lambda_T^h s^h \right), \quad (5.11)$$

where the teacher’s head coefficients $\lambda_T \in \Delta_{H-1}$ are $\lambda_T = \text{NORMALIZE}(\phi_T)$ with $\phi_T \in \mathbb{R}^H$.

In this formulation, $s^h \in \mathbb{R}^L$ represents the average *unnormalized attention logits* over all input elements, which are then combined according to λ_T and normalized with SOFTMAX to produce a distribution in Δ_{L-1} . We apply a normalization function NORMALIZE to head coefficients involved to create a *convex* combination over all heads in all layers. In this work we consider the sparse projection function NORMALIZE = SPARSEMAX [\[MA16\]](#), as:

$$\text{SPARSEMAX}(z) = \underset{p \in \Delta_{H-1}}{\text{argmin}} \|p - z\|_2.$$

We choose SPARSEMAX due to its benefits in terms of interpretability, since it leads to many heads having zero weight. We also found it outperformed every other projection we tried. [Figure 5.2](#) illustrates each step of our parameterized attention explainer.

5.5 Experiments

To evaluate our framework, we attempt to learn explainers for transformer models trained on three different tasks: text classification ([subsection 5.5.1](#)), image classification ([subsection 5.5.2](#)), and machine translation quality estimation (a text-based regression task,

detailed in subsection 5.5.3). We use JAX [Bra+18] to implement the higher-order differentiation, and use pretrained transformer models from the Huggingface Transformers library [Wol+20], together with Flax [Hee+20]. For each task, we train a teacher model with AdamW [LH19] but, as explained in section 5.3, we use SGD for the student model (inner loop). We also use scalar mixing [Pet+18] to pool representations from different layers automatically.¹ We train students with a teacher explainer in three settings:

- **No Explainer:** No explanations are provided, and no explanation regularization is used for training the student (i.e. $\beta = 0$ in Equation 5.3). We refer to students in this setting as **baseline** students.
- **Static Explainer:** Explanations for the teacher model are extracted with five commonly-used saliency-based explainers: (1) L2 norm of gradients; (2) a *gradient* $\times$ *input* explainer [DDF14]; (3) an *integrated gradients* explainer [STY17b]; and *attention* explainers that uses the *mean* pooling over attention from (4) all heads in the model and (5) from the heads of the last layer [FSA21; Vaf+21].
- **Learned Explainer (SMaT):** Explanations are extracted with the explainer described in section 5.4, with coefficients for each head that are trained with **SMaT** jointly with the student. We initialize the coefficients such that the model is initialized to be the same as the *static* attention explainer (i.e., performing the mean over all heads).

Independently of the T -explainer, we always use a learned attention-based explainer as the S -explainer, considering all heads except when the T -explainer is a static attention explainer that only considers the last layers’ heads, where we do the same for the S -explainer. We use the Kullback-Leibler divergence as $\mathcal{L}_{\text{expl}}$, and we set $\beta = 5$ for attention-based explainers and $\beta = 0.2$ for gradient-based explainers (since we found smaller values to be better). We set $\mathcal{L}_{\text{sim}}$ as the cross-entropy loss for classification tasks, and as the mean squared error loss for text regression. For each setting, we train five students with different seeds. Since there is some variance in students’ performance (we hypothesize due to the small training sets) we report the **median** and **interquantile range (IQR)** around it (relative to the 25-75 percentile).

5.5.1 Text Classification

For text classification, we consider the IMDB dataset [Maa+11], a binary sentiment classification task over highly polarized English movie reviews. As the base pretrained model, we use the small ELECTRA model [Cla+20], with 12 layers and 4 heads in each (total 48 heads).

Like the setting in [Pru+20a], we use the original training set with 25,000 samples to train the teacher, and further split the test set into a training set for the student and a dev and test set. We vary the number of samples the student is trained on between 500, 1,000, and 2,000. We evaluate *simulability* using accuracy (i.e., what percentage of student

¹While scalar mixing reduced variance of student performance, SMaT also worked with other common pooling methods.

Table 5.1: Results for the IMDB dataset with respect to student *simulability* in terms of accuracy (%). *Underlined* values indicate higher simulability than baseline with non-overlapping IQR.

	500	1,000	2,000
No Explainer	81.72 [81.24:81.75]	83.44 [83.36:83.63]	84.84 [84.80:84.88]
Gradient L2	81.66 [81.32:82.00]	82.98 [82.72:83.08]	84.78 [84.96:85.08]
Gradient $\times$ Input	<u>84.83</u> [84.79:84.88]	81.15 [80.95:81.36]	83.84 [83.59:84.99]
Integrated Gradients	<u>82.99</u> [82.59:82.99]	81.79 [81.72:81.87]	84.20 [84.03:85.03]
Attention (<i>all layers</i>)	<u>83.00</u> [82.60:83.00]	<u>85.72</u> [85.72:86.23]	<u>90.08</u> [89.72:90.11]
Attention (<i>last layer</i>)	80.91 [79.99:81.07]	83.15 [82.91:83.51]	<u>91.47</u> [91.39:91.56]
Attention (SMaT)	<u>91.48</u> [91.40:91.56]	<u>92.56</u> [92.28:92.83]	<u>92.84</u> [92.84:93.08]

predictions match with teacher predictions). The teacher model obtains 91% accuracy on the student test set.

Table 5.1 shows the results in terms of simulability (Equation 5.1) for the three settings. We can see that, overall, the attention explainer trained with SMaT leads to students that simulate the teacher model much more accurately than students trained without any explanations, and more accurately than students trained with any *static* explainer across all student training set sizes. Interestingly, the gradient-based explainers only improve over the baseline students when the amount of training data is very low, and actually degrade simulability for larger amounts of data. Using only heads from the last layer seems to have the opposite effect, leading to higher simulability than all other static explainers only for larger training sets.

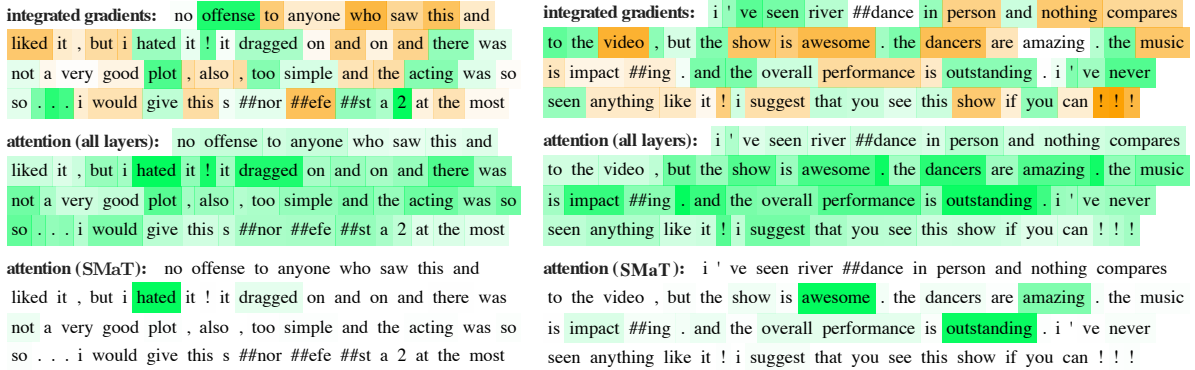


Figure 5.3: Explanations given by integrated gradients, attention (*last layer*), and our learned attention explainer (SMaT) for two movie reviews of the IMDB dataset (negative and positive examples). Green and orange represent positive and negative contributions, respectively.

Plausibility analysis. We select the median model trained with 1,000 samples and extract explanations for test samples from the MovieReviews dataset [DeY+20], which contains binary sentiment movie reviews from Rotten Tomatoes alongside human-rationale annotation. Since the labels are binary (indicating whether a token is part of the explana-

	AUC
Gradient L2	0.65
Gradient $\times$ Input	0.51
Integrated Gradient	0.53
Attention (<i>all layers</i>)	0.68
Attention (<i>last layer</i>)	0.61
Attention (SMaT)	0.73
Attention (<i>best layer</i>)*	0.75
Attention (<i>best head</i>)*	0.75

Table 5.2: *Plausibility* on *MovieReviews* in terms of AUC. * represents methods that use human labels.

tion or not) and the predicted scores are real values, we follow [Fom+21] and report our results in terms of the Area Under the Curve (AUC), which automatically considers multiple binarization thresholds. The results are shown in Table 5.2 along with two randomly selected examples of extracted explanations in Figure 5.3. We found that gradient-based explanations are less plausible than those using attention (with the exception of *Grad. L2*, which is similar to static attention) and that ones produced with SMaT achieve the highest plausibility, indicating that our learned explainer can produce human-like explanations while maximizing simulability. Moreover, SMaT achieves a similar AUC score to the best performing attention layer and head,² while not requiring *any* human annotations. This is evidence that scaffolded simulability, while not explicitly designed for it, is a good proxy for plausibility and “human-like” explanations.

5.5.2 Image Classification

To validate our framework across multiple modalities, we consider image classification on the CIFAR-100 dataset [Kri09]. We use as the base model the Vision Transformer (ViT) [Dos+20], in particular the base version with 16×16 patches that was only pretrained on ImageNet-21k [Rid+21]. We up-sample images to to a 224×224 resolution.

Since the self-attention mechanism in the ViT model only works with patch representations, the explanations produced by attention-based explainers will be at patch-level rather than pixel-level. We split the original CIFAR-100 training set into a new training set with 45,000 and a validation set with 5,000. Unlike the previous task, we reuse the training set for both the teacher and student, varying the number of samples the student is trained with between 2,250 (5%), 4,500 (10%) and 9,000 (20%). We use accuracy as the simulability metric and the teacher obtains 89% on test set.

Table 5.3 shows the results for the three settings. Similarly to the results in the text modality, the attention explainer trained with SMaT achieves the best scaffolding performance, although the gaps to static attention-based explainers are smaller (especially

²AUC scores obtained by independently trying all attention heads and layers of the model.

Table 5.3: *Simulability* results, in terms of accuracy (%), on the CIFAR100 dataset. *Underlined* values represent better performance than baseline with non-overlapping IQR

	2,250	4,500	9,000
No Explainer	81.16 [80.98:81.26]	84.02 [83.98:84.24]	85.20 [85.17:85.26]
Gradient L2	80.97 [80.91:81.10]	83.98 [83.81:84.23]	85.13 [84.97:85.50]
Gradient $\times$ Input	80.93 [80.82:81.04]	83.99 [83.98:84.13]	85.33 [84.85:85.35]
Integrated gradients	80.22 [80.17:80.35]	83.44 [83.25:83.44]	84.99 [84.76:85.22]
Attention (<i>all layers</i>)	<u>82.53</u> [82.53:82.62]	<u>84.81</u> [84.74:84.92]	<u>85.92</u> [85.78:85.94]
Attention (<i>last layer</i>)	<u>82.34</u> [82.30:82.60]	<u>84.65</u> [84.56:84.81]	85.31 [84.84:85.31]
Attention (SMaT)	<u>83.09</u> [82.77:83.28]	<u>85.42</u> [85.39:85.85]	<u>85.96</u> [85.74:86.35]

when students are trained with more samples). Here, the gradient-based explainers always degrade simulability across the tested training set sizes and and it seems important that the explanations include attention information from layers other than the last one.

Plausibility analysis. Since there are no available human annotations for plausibility in the CIFAR-100 dataset, we design a user study to measure the plausability of the considered methods. The original image and explanations extracted with Gradient $\times$ Input, Integrated Gradients, Attention (*all layers*), and Attention (SMaT) are shown to the user, and the user has to rank the different explanations to answer the question “*Which explanation aligns the most with how you would explain a similar decision?*”. Explanations were annotated by three volunteers. After collecting results, we compute the *rank* and the *TrueSkill* rating [HMG07] for each explainer (roughly, the “skill” level if the explainers where players in game). The results are shown in Table 5.4. As in the previous task, attention trained with SMaT outperforms all other explainers in terms of plausibility, and its predicted *rating* is much higher than all other explainers. We also show examples of explanations for a set of randomly selected images in Figure 5.4.

Rank	<i>TrueSkill</i>	
Grad. $\times$ Input	3-4	-2.7 $\pm$.67
Integ. Grad.	3-4	-2.1 $\pm$.67
Attn. (<i>all lx.</i>)	2	0.7 $\pm$.67
Attn. (SMaT)	1	4.3$\pm$.70

Table 5.4: *Plausibility* results of the human study on visual explanations.

5.5.3 Machine Translation Quality Estimation

Quality Estimation (QE) is the task of predicting a quality score given a sentence in a source language and a translation in a target language from a machine translation system, which requires models that consider interactions between the two inputs, source and target. Scores tend to be continuous values (making this a regression task) that were collected from expert annotators.

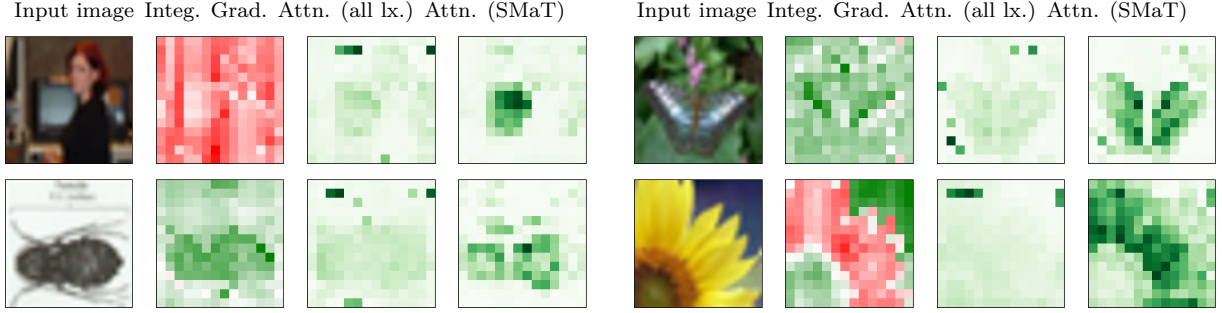


Figure 5.4: Explanations given by integrated gradients, attention (all layers), and learned attention explainer for a set of input images of CIFAR-100. Gold labels are: “television”, “butterfly”, “cockroach”, and “sunflower”.

Table 5.5: *Simulability* results, in terms of Pearson correlation, on the ML-QE dataset. *Underlined* values represent better performance than baseline with non-overlapping IQR.

	2,100	4,200	8,400
No Explainer	.7457 [.7366:.7528]	.7719 [.7660:.7802]	.7891 [.7860:.7964]
Gradient L2	<u>.8065</u> [.8038:.8268]	<u>.8535</u> [.7117:.8544]	.8638 [.8411:.8657]
Gradient $\times$ Input	.6846 [.6781:.6894]	.6922 [.6885:.6965]	.7141 [.7136:.7147]
Integrated gradients	.6686 [.6677:.6694]	.7086 [.6994:.7101]	.7036 [.6976:.7037]
Attention (<i>all layers</i>)	<u>.8120</u> [.7955:.8125]	<u>.8193</u> [.8186:.8280]	.8467 [.8464:.8521]
Attention (<i>last layer</i>)	.7486 [.7484:.7534]	.7720 [.7672:.7726]	.7798 [.7717:.7814]
Attention (SMaT)	.8156 [.8096:.8183]	.8630 [.8412:.8724]	.8561 [.8512:.8689]

Interpreting quality scores of machine translated outputs is a problem that has received recent interest [Fom+21] since it allows identifying which words were responsible for a bad translation. We use the MLQE-PE dataset [Fom+20], which contains 7,000 training samples for each of seven language pairs alongside word-level human annotation. We use as the base model a pretrained XLM-R-base [Con+19], a multilingual model with 12 layers and 12 heads in each (total of 144 heads).

We exclude one of the language pairs in the dataset (**si-en**) since the XLM-R model did not support it, leading to a training set with 42,000 samples. Similar to the CIFAR-100 case, we reuse the same training set for both the teacher and student, sampling a subset for the latter. We vary the number of samples the student is trained with between 2,100 (5%), 4,200 (10%) and 8,400 (20%). Since this is a regression task, we evaluate simulability using the Pearson correlation coefficient between student and teacher’s predictions.³ The teacher achieves 0.63 correlation on the test set.

Table 5.5 shows the results for the three settings. Similar to other tasks, the attention explainer trained with SMaT leads to students with higher simulability than baseline students and similar or higher than *static* explainer across all training set sizes. Curiously, the *Grad. L2* explainer achieves very high simulability for this task. It even has a higher

³Pearson correlation is the standard metric used to evaluate sentence-level QE models.

Table 5.6: Plausibility results for source and target inputs for each language pair of the MLQE-PE dataset in terms of AUC. * represents *supervised* methods that use human labels in some form.

	EN-DE		EN-ZH		ET-EN		NE-EN		RO-EN		RU-EN		OVERALL	
	src.	tgt.	src.	tgt.	src.	tgt.	src.	tgt.	src.	tgt.	src.	tgt.	src.	tgt.
Gradient L2	0.64	0.65	0.65	0.49	0.67	0.61	0.68	0.55	0.72	0.68	0.65	0.54	0.67	0.59
Gradient $\times$ Input	0.58	0.60	0.61	0.51	0.60	0.54	0.61	0.49	0.64	0.59	0.58	0.51	0.61	0.54
Integrated Gradients	0.59	0.60	0.63	0.49	0.60	0.52	0.64	0.48	0.64	0.59	0.60	0.51	0.62	0.53
Attention (<i>all layers</i>)	0.60	0.63	0.68	0.52	0.60	0.61	0.58	0.55	0.66	0.70	0.62	0.55	0.62	0.59
Attention (<i>last layer</i>)	0.51	0.49	0.61	0.49	0.51	0.50	0.55	0.48	0.52	0.57	0.56	0.50	0.54	0.50
Attention (SMaT)	0.64	0.65	0.68	0.52	0.66	0.64	0.66	0.54	0.71	0.70	0.61	0.54	0.66	0.60
Attention (<i>best layer</i>)*	0.64	0.65	0.69	0.64	0.64	0.68	0.68	0.68	0.71	0.76	0.64	0.59	0.65	0.65
Attention (<i>best head</i>)*	0.67	0.67	0.70	0.65	0.70	0.70	0.70	0.69	0.73	0.75	0.67	0.60	0.67	0.66

median simulability score than SMaT for 8,400 samples. However, we attribute this to variance in the student training set sampling (that could lead to an imbalance in language pair proportions) which could explain why SMaT performance degrades with more samples. For this task, the gradient-based explainers always degrade simulability across the tested training set size. It also seems that using only the last layer’s attention is also ineffective at teaching students, achieving the same performance as the baseline.

Plausibility analysis. We select the median model trained with 4,200 samples and follow the approach devised in the Explainable QE shared task to evaluate plausibility [Fom+21], which consists of evaluating the human-likeness of explanations in terms of AUC only on the subset of translations that contain errors. The results are shown in Table 5.6. We note that for all language pairs, SMaT performs on par or better than static explainers, and only being surpassed by *Grad. L2* in the *source-side* over all languages. Comparing with the best attention layer/head, an approach used by Fomicheva et al. [FSA21] and Treviso et al. [Tre+21], SMaT achieves similar AUC scores for source explanations, but lags behind the best attention layer/head for target explanations on *-EN language pairs. However, as stressed previously for text and image classification, SMaT sidesteps human annotation and avoids the cumbersome approach of independently computing plausibility scores for all heads.

5.6 Related Work

Explainability for text & vision. Several works propose explainability methods to interpret decisions made by NLP and CV models. Besides gradient and attention-based approaches already mentioned, some extract explanations by running the models with perturbed inputs [RSG16; Fen+18; Kim+20]. Others even define custom backward passes to assign relevance for each feature [Bac+15]. These methods are commonly employed together with post-processing heuristics, such as selecting only the top-k tokens/pixels with higher scores for visualization. Another line of work seeks to build a classifier with inherently interpretable components, such as methods based on attention mechanisms and

rationalizers [LBJ16; BAT19].

Evaluation of explainability methods. As mentioned in the introduction, early works evaluated explanations based on properties such as *consistency*, *sufficiency* and *comprehensiveness*. Jacovi and Goldberg [JG20] recommended the use of a graded notion of faithfulness, which the ERASER benchmark quantifies using the idea of sufficient and comprehensive rationales, alongside compiling datasets with human-annotated rationales for calculating plausibility metrics [DeY+20]. Given the disagreement between explainability methods, Neely et al. [Nee+21] showed that without a faithful ground-truth explanation it is impossible to determine which method is better. Diagnostic tests such as the ones proposed by Adebayo et al. [Ade+18] and Wiegrefe and Pinter [WP19] and Atanasova et al. [Ata+20] are more informative yet they do not capture the main goal of an explanation: the ability to communicate an explanation to a practitioner.

Simulability. A new dimension for evaluating explainability methods relies on the forward prediction/simulation proposed by Lipton [Lip16a] and Doshi-Velez and Kim [DK17], which states that humans should be able to correctly simulate the model’s output given the input and the explanation. Chandrasekaran et al. [Cha+18], Hase and Bansal [HB20], and Arora et al. [Aro+22] analyze simulability via human studies across text classification datasets. Treviso and Martins [TM20] designed an automatic framework where students (machine or human) have to predict the model’s output given an explanation as input. Similarly, Pruthi et al. [Pru+20a] proposed the simulability framework that was extended in our work, where explanations are used to regularize the student rather than passed as input.

Learning to explain. The concept of simulability also opens a path to learning explainers. In particular Treviso and Martins [TM20] learn an attention-based explainer that maximizes simulability. However, directly optimizing for simulability sometimes led to explainers that learned trivial protocols (such as selecting only punctuation symbols or stopwords to leak the label). Our approach of optimizing a teacher-student framework is similar to approaches that optimize for model distillation [ZXM21]. However, these approaches modify the original model rather than introduce a new explainer module. Raghu et al. [Rag+21] propose a framework similar to ours for learning *commentaries* for inputs that speed up and improve the training of a model. However commentaries are model-independent and are optimised to improve performance on the real task. Rationalizers [Che+18; JG21; GM21] also directly learn to extract explanations, but can also suffer from trivial protocols.

5.7 Conclusion & Subsequent Work

We proposed **SMaT**, a framework for directly optimizing explanations of the model’s predictions to improve the training of a student *simulating* the said model. We found that, across tasks and domains, explanations learned with SMaT both lead to students that simulate the original model more accurately and are more aligned with how people explain similar decisions when compared to previously proposed methods. On top of that, our parameterized attention explainer provides a principled way for discovering relevant

attention heads in transformers.

Our work shows that scaffolding/simulability is a suitable criterion for both evaluating and optimizing explainability methods. Crucially, we believe that there are further unexplored connections between explainability and model distillation: for instance, subsequent work has shown *chain-of-thought* rationales extracted from large LLMs can be used to improve the training of smaller models [Li+23], which means that these can indeed be *good* explanations for models' decisions.

Part III

Learning with AI Teachers (**Proposal**)

Chapter 6

Learning Signals from AI Teachers

The increasing capabilities of LLMs have made viable their use for providing reliable feedback on other models' outputs (as we saw in [chapter 4](#)), and this feedback has been increasingly used to align and improve the performance of student models ([subsection 2.2.4](#)). However, as we saw in the previous chapter, feedback is not the only learning signal we can extract from AI models: explanations for outputs can also be used to improve the performance of student models. Additionally, recent work has also shown that LLMs can also be used to generate *synthetic data* which can align models better than real data or human feedback [[Zho+23](#)]. However, these different *learning signals* that can be extracted from *AI teachers* have not yet been compared systematically in terms of their effectiveness at aligning and teaching student models.

In this chapter, we propose unifying feedback, explanations, and synthetic data as learning signals from AI teachers, which can be used to align (pre-trained) models and/or improve their performance at a task. In particular, we propose using the task of Machine Translation to study this and using *multilingual* LLMs to provide feedback for student models' translations, explanations for translations, or synthetic data/translations for training.

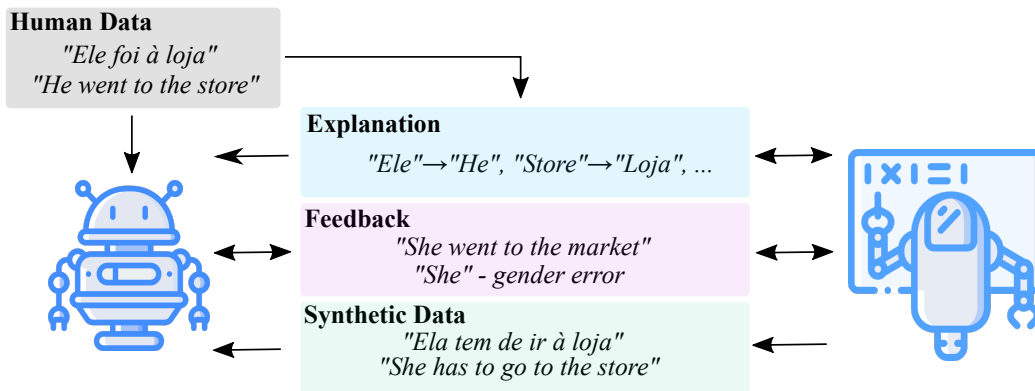


Figure 6.1: Illustration of the different (example) learning signals that can be extracted from AI teachers for Machine Translation

6.1 Unifying Learning Signals

We first consider a (student) model S , a (conditional) language model potentially already pretrained on some dataset. With access to some supervised finetuning (ST) dataset $\mathcal{D}$ (for some specific task or instruction following), we can first consider training this model using standard maximum likelihood estimation (Equation 2.6).

Now, we consider we also have some teacher model T , a (conditional) language model as well that is already well *aligned* to perform the task (by, for example, having already been supervised-finetuned and/or seen feedback) and from which that can extract three types of learning signals:

1. **Synthetic Data**, by generating an example output for a particular input

$$T_d : \mathcal{X} \rightarrow \mathcal{Y} \quad \text{or} \quad T_d : \mathcal{C} \rightarrow \mathcal{X} \times \mathcal{Y}$$

2. **Feedback** on one or more (model) outputs given an input

$$T_f : \mathcal{X} \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_n \rightarrow \mathcal{F}$$

3. **Explanations** for a particular output¹ given an input

$$T_e : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{E}$$

For example, given a general-purpose LLM (that has been *instruction-tuned*), these different learning signals can be extracted by *prompting* it to provide the specific signal (as in chapter 4, or for *chain-of-thought* explanations) but could also be extracted from some internal state of the models (like in chapter 5). Additionally, these signals can also be conditioned on some extra context $c \in \mathcal{C}$, for example, by asking for new synthetic data with particular properties.

The learning signals can be integrated as part of the training/alignment of the student model: for synthetic data (1), we can simply add the generated examples to the training dataset, potentially weighting them differently or using some form of curriculum learning; for feedback (2), we can collect a set of input-predicted output pairs $\hat{\mathcal{D}}$, collect feedback from the teacher model on these pairs, and use this to optimize the student model (subsection 2.2.2); for explanations, we can collect explanations for the real outputs in the training dataset, and integrating these as an auxiliary loss in the student training (like in chapter 5, or by distilling free-form *rationales* [Li+23]).

Now, to compare the effectiveness of these different learning signals, we can train different student models using each of these signals, and evaluate their performance on a held-out test set. To ensure a fair comparison, all student models should start from the same pretrained model and have access to the same amount of real data. Importantly, we also have to compare the *cost* of each learning signal: simple feedback formats (like numerical scores) can be extracted from LLMs at a lower cost than, say, synthetic data (as the latter requires more *decoding* steps). Comparing the effectiveness to adding more real data (or even actual human feedback) is also something we are planning to explore.

¹In general, explanations are obtained for correct or aligned outputs, while feedback is for (potentially) incorrect or unaligned model outputs, but this distinction is loose.

6.2 Proposed Experiments

To study and compare the effectiveness of the different learning signals, we propose starting to explore how these might help in aligning pretrained conditional language models for the task of Machine Translation. We propose using multilingual LLMs, for both the student & teacher, as some of these have shown to perform well at MT (even without additional training) [Vil+22; Xu+23a] and their general-purpose nature makes them suitable for easily extracting different learning signals. However, for the student model, we plan on using *raw*, *base* LLMs which have only undergone some form of multilingual pretraining and haven't been explicitly aligned for MT, while for the teacher model, we plan on using LLMs already explicitly aligned for MT, and that are potentially larger and/or more capable. We plan on studying varying sizes and model families for both the student and teacher models.

Then, leveraging some existing human-generated datasets for MT (for example, from OPUS [TT20]), we plan on comparing the effectiveness of training these students only on the real data vs training them also using the different learning signals extracted from the teacher model: for example, we could further apply reinforcement learning using translation quality scores obtained from the teacher model (chapter 4). We could compare effectiveness by evaluating student models using state-of-the-art metrics for MT (like COMET [Rei+20a]) and potentially human evaluation.

6.3 Research Questions

Our research aims to answer the following questions, thus providing insights that could steer the future usage of AI teachers for student model training:

1. *Which types of feedback and explanations are most effective at teaching a student model?* While it is generally recognized that *AI feedback* can be used to improve student models, most work has focused on very simple feedback formats, rather than exploring more integrating more complex feedback formats in the training (like error spans produced with the AutoMQM prompt). Additionally, the effectiveness of explanations has not been systematically compared to feedback.
2. *Does the effectiveness of feedback, explanations and synthetic data depend on the capacity of the student model or the capabilities of the teacher model?* It is conceivable that more complex student models can better utilize certain types of learning signals. Similarly, teacher models with varying capabilities may excel in providing different types of feedback and explanations: we saw in chapter 4 that while all LLMs explored were able to provide numerical feedback, only the largest ones were able to provide decent fine-grained annotations.
3. *What strategies for integrating feedback, explanations and/or synthetic data into the student model's learning process lead to more aligned models?* In the same way that it has been found that supervised finetuning followed by human feedback leads to better alignment, the exact order and way in which AI learning signals are integrated into the student model's learning process may also affect the alignment of the student model.

4. *Can learning signals from AI teachers that are weaker than the student (for example, having fewer parameters) still be used to align and improve the latter?* Since the teacher has already been aligned for the task of interest, it's possible that even if it's weaker than the student, it could provide beneficial learning signals for alignment [Bur+23], and findings could be used for the next chapter.

Multi-Generational Training of Language Models

Currently, most language models are (pre-)trained on large amounts of data *from scratch*, with each new model being trained independently of the knowledge learned by previous models and only indirectly using some “*meta-information*” from their training (such as the choice of training data, hyperparameters, etc). However, this process is extremely inefficient in the amount of information it wastes since all information encoded weights. This paradigm leads to the need for ever-increasing amounts of data to make up for this inefficiency and further improve models, an approach that might soon reach its limits as we exhaust the amounts of readily available human-produced data.

Recent work has shown that we can indeed leverage information from trained models to improve the training of new ones by, for example, using them to select good-quality (human-written) training data [Sac+24] or even re-writing the training data to be of higher-quality (i.e. generate synthetic data) [Mai+24]. However, these approaches have only been explored in isolation and without a clear understanding of how to best leverage the knowledge of previous models to train new ones. Additionally, it is currently unclear if these approaches can be applied iteratively across multiple *generations* of models.

In this chapter, we propose a new paradigm for training language models, where models are trained in a *multi-generational* manner, with each new generation of models learning from the previous (weaker) generation’s knowledge. Using the findings from the previous chapter, we aim to use AI teachers’ signals - including feedback, explanations and synthetic data - to train subsequent generations of models, thereby reducing the overall amount of *human-written* data required and also the computational resources (by training models that have fewer parameters or were trained for fewer steps). Unlike the previous chapter, however, we plan on not only focusing on aligning student models that have already been pretrained but also studying how these learning signals can be leveraged during the first pre-training phase, as it represents the bulk of the resources used in large model training.

7.1 Proposed Experiments

To test and explore the viability of multi-generational model training, we propose a series of experiments around training general-purpose multilingual LLMs, which can later be further aligned for specific tasks such as MT.

- Initially, we would train a baseline, *first-generation* LLMs using conventional pre-training methods on a large dataset, such as mC4. Optionally, we could also already start with an existing, moderate-capacity trained LLM.
- Then, with access to the same human-written data, would train a new generation model using the previous one as an *AI Teacher*: we could extract explanations/feed-back for the whole pretraining set (for augmenting pre-training loss or dataset filtering) or generate large amounts of synthetic pretraining data. The teacher could also be used online during the pretraining process, to provide feedback on the model’s outputs. Optionally, the teacher could also first be trained to provide better learning signals for the student model.
- Repeat the process if the new generation model displays improved performance over the previous generation model.

At each generation, we would evaluate not only at specific tasks such as MT but also on more general benchmarks. Crucially, what distinguishes this setting from *distillation* is that want our student model to get better than its teacher. This means the new generation student might be larger than its teacher (but not necessarily). We also have to take into account the computational cost of training and using the teacher model and compare it to the cost of training the student model without the teacher.

7.2 Research Questions

1. *Which learning signals from AI teachers are most effective for the initial pre-training of student models?* The previous chapter focused on finding the best setting to use AI teachers for *aligning* pretrained models for a specific task, but this setting might change when considering different learning dynamics of the pre-training phase.
2. *When training student models of the same size as the teacher, can train equal or better students more sample-efficiently (compared to the teacher)?* Data is one of the most scarce resources for training LLMs, so improving the sample-efficiency of model training would allow us to push the currently existing human data much further.
3. *When training student models larger than the teacher, can we match or surpass models trained without any teacher in a compute-efficient matter?* Increasing model size by itself already improves the performance when training on the same data (without teachers), but learning signals from the teacher could further improve the *scaling behaviour* of the student model.
4. *Can we leverage learning signals from multiple AI teachers to improve the student model? How does this compare to using a single teacher?* This question is particularly

relevant if the student model is larger than a single teacher, and where benefits from its learning signals could be limited due to the capability gap. In this case, using multiple teachers (with complementary strengths and weaknesses) might help, but brings questions on how to combine their learning signals.

5. *Can we iteratively improve models across multiple generations? Are there diminishing returns with each generation? How does relate the capability gap between teacher(s) and student?* As we progress through generations, the knowledge of the teacher models might become less useful or existing biases might be amplified too much, leading to learning signals being actively harmful to the performance.

Part IV

Conclusions

Conclusions and Future Directions

In this proposal, I described my finished work on leveraging *feedback* models to improve and align Machine Translation systems ([Chapter 3](#)) and explored how large language models (LLMs) can be used to provide (fine-grained) feedback, minimizing the need for human annotations ([Chapter 4](#)). I also described my finished work on evaluating and improving *explanations* of neural models’ decisions by using them to teach other models ([Chapter 5](#)). I then proposed a new research direction to systematically compare the effectiveness of different *learning signals*, such as feedback, explanations and synthetic data, by using LLMs as AI teachers ([Chapter 6](#)), further proposed a *multi-generational* framework for iteratively training and improving new generations of language models using learning signals from models in previous generations ([Chapter 7](#)).

In the final thesis document, this section will include a more complete summary of the takeaways of the entire thesis, as well as my perspective on future directions in the field.

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